

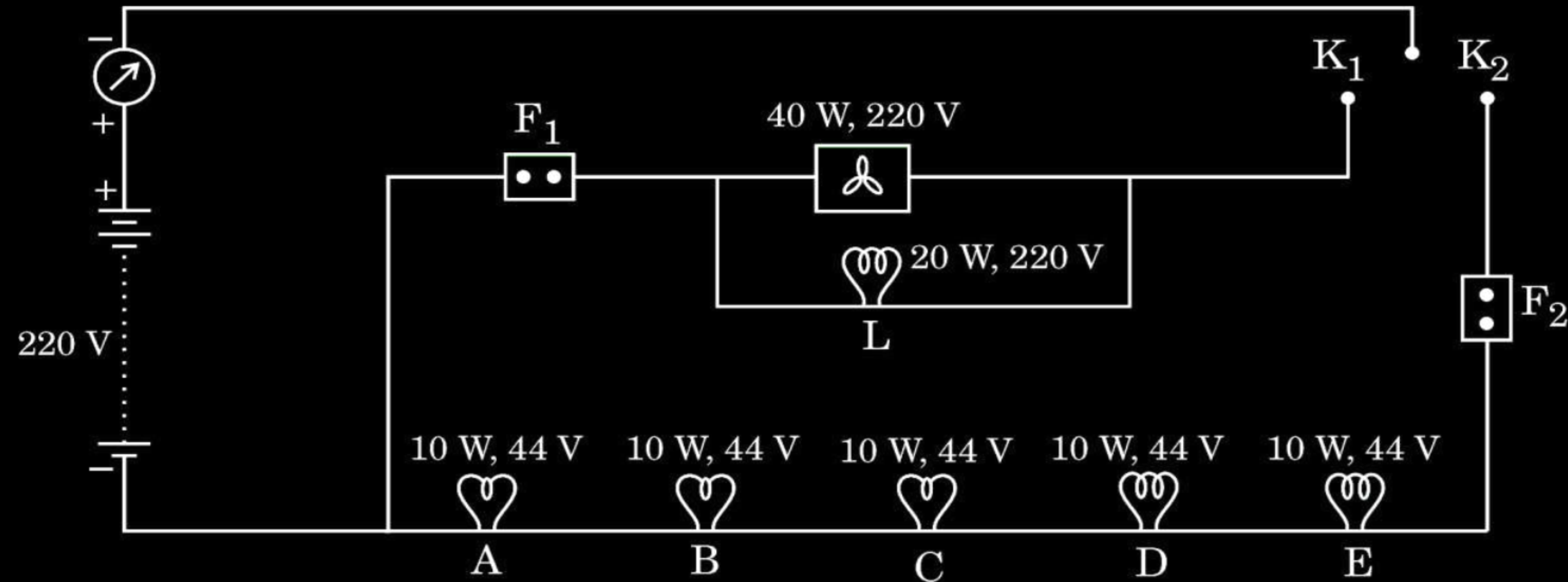
31. (a) Write mathematical expression for Joule's law of heating.

(b) Calculate the amount of heat generated while transferring 72000 coulomb of charge in two hours through a potential difference of 220 V.

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38. In a domestic circuit, six LED bulbs and a ceiling fan are arranged as per the circuit shown in the figure. The source voltage is 220 volts and the power rating of each device is marked in the circuit diagram. Two fuses F_1 and F_2 are also provided in the circuit. Based on the circuit diagram, answer the following questions :

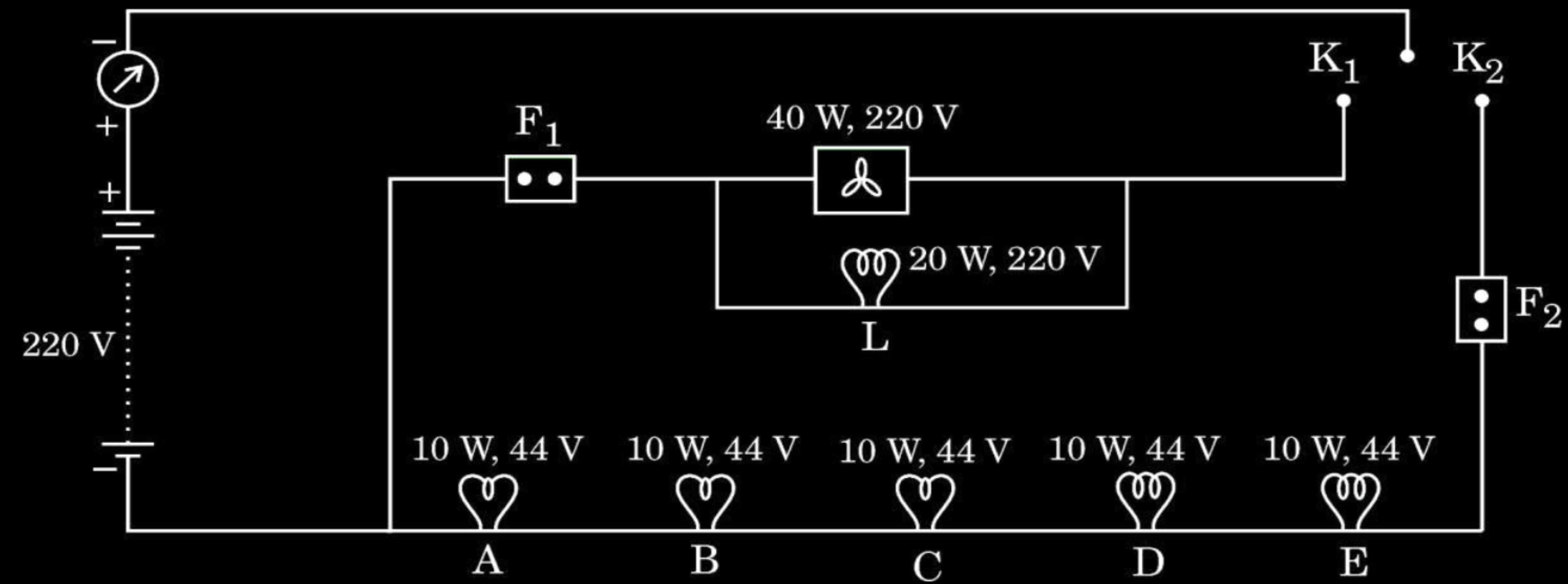
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- (a) State what happens when 1
- (i) only key K_1 is closed,
 - (ii) and key K_2 is also closed.
- (b) Find the current drawn by the bulb L. 1
- (c) (i) Calculate the resistance of the bulb A. Find the total resistance offered by the circuit to the flow of current when only key K_2 is closed. 2

OR

- (c) (ii) What would happen to the glow of the bulbs in the circuit when K_1 and K_2 both are closed and the fuse F_1 melts 2



- 36.** (a) (i) If R_s and R_p respectively are the total resistances of the series and parallel combinations of the three resistors of resistances R_1 , R_2 and R_3 , then write the relationships among (I) R_s , R_1 , R_2 and R_3 and (II) R_p , R_1 , R_2 and R_3 .
- (ii) An electric oven operates on the mains voltage of 220 V. This oven consumes 2.2 kWh of electrical energy in one hour. Calculate :
- (I) power rating of the oven,
 - (II) current drawn by the oven, and
 - (III) resistance of the oven when in operation.

- (b) (i) Define the term electric power. Express it in terms of potential difference (V) and resistance (R).
- (ii) The resistance of a metal wire of length 6 m is $120\ \Omega$. If the area of cross-section of the wire is $4 \times 10^{-7}\ \text{m}^2$, calculate the electrical resistivity of the material of the wire. How would the resistivity be affected if this wire is gradually stretched so that its length is doubled ? Justify your answer.

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11. An electric bulb is connected to a power supply of 220 V. If the current drawn by the bulb from the supply is 500 mA, the power of the bulb is :

(A) 11 W

(B) 110 W

(C) 220 W

(D) 1100 W

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11. An electric bulb is connected to a power supply of 220 V. If the current drawn by the bulb from the supply is 500 mA, the power of the bulb is :

(A) 11 W

(B) 110 W

(C) 220 W

(D) 1100 W

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(B)/ 110 W

(b) Define electric power. When do we say that the power consumed in an electric circuit is 1 watt ?

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(b) Define electric power. When do we say that the power consumed in an electric circuit is 1 watt ?

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2

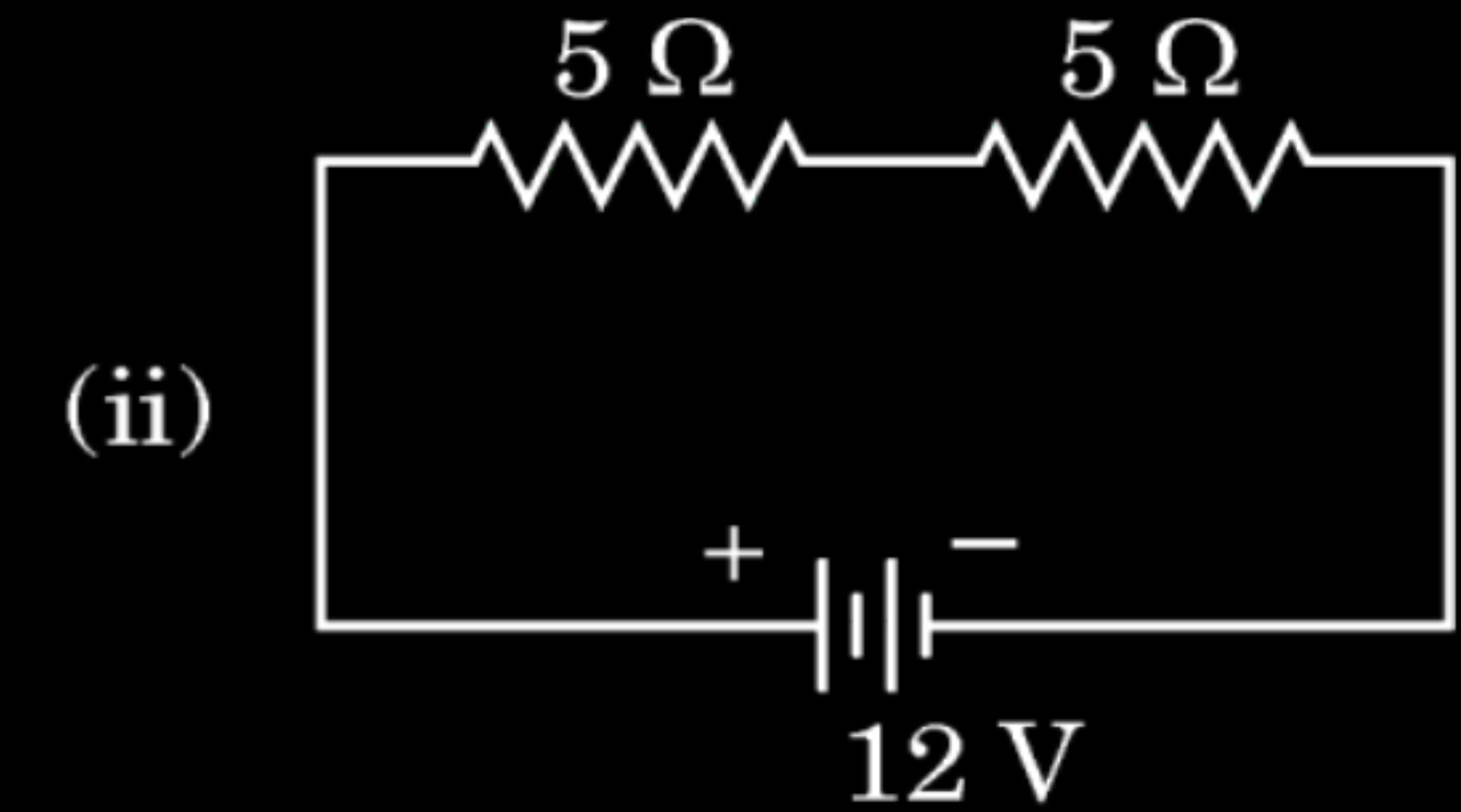
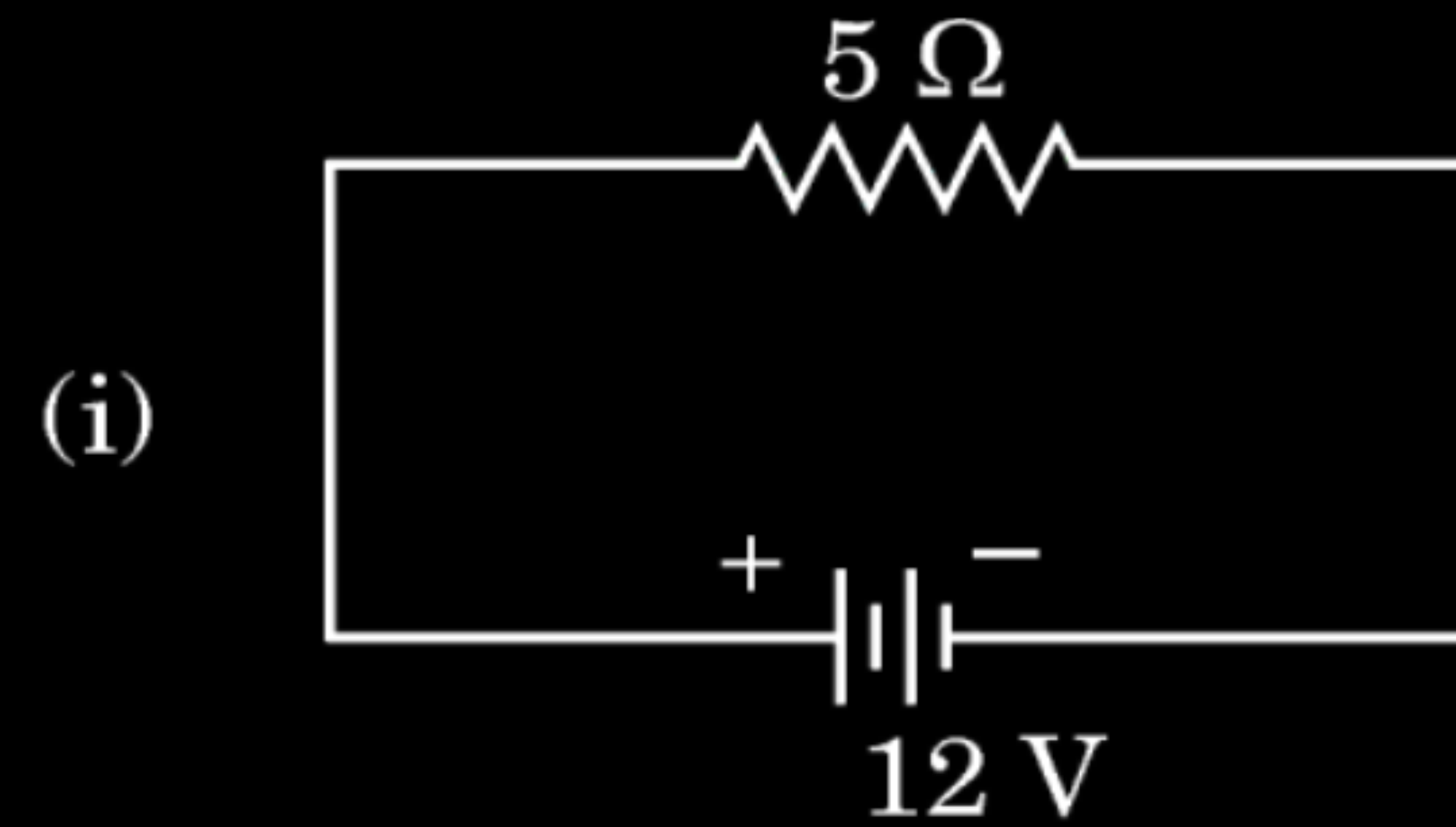
(b)

- Electric power is the rate at which electrical energy is consumed or dissipated in an electrical circuit. / Electric power is electrical energy consumed per unit time.
- When 1A current at 1V potential difference flows in a circuit. / Power consumed is 1W when 1J electrical energy is consumed in 1 s.

1

1

22. (a) Consider the following circuits :



In which circuit will the power dissipated in the circuit be
(I) minimum (II) maximum ? Justify your answer.

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• (a) Power dissipated will be: • (I) minimum in circuit (ii) and • (II) maximum in circuit (iii) • • Justification : Power dissipated = $\frac{V^2}{R}$ • Since voltage is same in three cases, the power dissipation • will be inversely proportional to the resistance of the • circuit. • • (Award marks if explained through calculations)	$\frac{1}{2}$ $\frac{1}{2}$ 1
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- (b) Two lamps, rated 100 W; 220 V and 60 W; 220 V are connected in parallel to electric main supply of 220 V. Find the current drawn by the two lamps from the supply.

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2

(b)

Voltage $V = 220\text{V}$

$$P = V \times I$$

Power $P_1 = 100\text{ W}$

$P_2 = 60\text{ W}$

$$\therefore I_1 = \frac{P_1}{V} = \frac{100\text{ W}}{220\text{ V}} = 0.45\text{ A}$$

$$\therefore I_2 = \frac{P_2}{V} = \frac{60\text{ W}}{220\text{ V}} = 0.27\text{ A}$$

$\therefore$ Total current drawn

$$I = I_1 + I_2 = 0.45 + 0.27 = 0.72\text{ A}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 22.** (a) Consider two lamps A and B of rating 50 W; 220 V and 25 W; 220 V respectively. Find the ratio of the resistances of the two lamps (i.e. $R_A : R_B$).

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22. (a) Consider two lamps A and B of rating 50 W; 220 V and 25 W; 220 V respectively. Find the ratio of the resistances of the two lamps (i.e. $R_A : R_B$).

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2

(a) $P = \frac{V^2}{R}$	$\frac{1}{2}$
$R = \frac{V^2}{P}$ $R_A = \frac{(220)^2}{50}$	
$R_B = \frac{(220)^2}{25}$	$\frac{1}{2}$
$\frac{R_A}{R_B} = \frac{1}{2}$	1
/	
$P = \frac{V^2}{R}$ Since V is same means P is inversely proportional to R.	
Power of lamp A is double that of B hence Resistance of A is half that of B.	1
$\frac{R_A}{R_B} = \frac{1}{2}$	1

- (b) Heat produced per second due to a current in a resistor of $4\ \Omega$ is 400 joules. Calculate the potential difference across the resistor. 2

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(b)

$$H = \frac{V^2}{R} \times t$$

$$400 = \frac{V^2}{4} \times 1$$

$$V^2 = 1600$$

$$V = 40\text{V}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

13. An electric bulb is rated 220 V; 11W. The resistance of its filament when it glows with a power supply of 220 V is :

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- (a) 4400 Ω
- (b) 440 Ω
- (c) 400 Ω
- (d) 20 Ω

13. An electric bulb is rated 220 V; 11W. The resistance of its filament when it glows with a power supply of 220 V is :

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1

- (a) 4400 Ω
- (b) 440 Ω
- (c) 400 Ω
- (d) 20 Ω

(a)/4400 Ω

14. The minimum number of identical bulbs of rating 4V; 6W, that can work safely with desired brightness, when connected in series with a 240 V mains supply is :

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(a) 20

(b) 40

(c) 60

(d) 80

14. The minimum number of identical bulbs of rating 4V; 6W, that can work safely with desired brightness, when connected in series with a 240 V mains supply is :

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1

- (a) 20
- (b) 40
- (c) 60
- (d) 80

(c)/60

- 26.** A voltage source sends a current of 2 A to a resistor of $40\ \Omega$ connected across it for 5 minutes. Calculate the electrical energy supplied by the source.

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- 26.** A voltage source sends a current of 2 A to a resistor of 40 Ω connected across it for 5 minutes. Calculate the electrical energy supplied by the source.

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Given $I = 2 \text{ A}$, $R = 40 \text{ } \Omega$, $t = 5 \text{ minutes} = 300 \text{ s}$

$$\text{Electrical energy} = I^2 R t$$

$$= (2 \text{ A})^2 \times 40 \text{ } \Omega \times 300 \text{ s}$$

$$= 48000 \text{ J}$$

$\frac{1}{2}$

1

$\frac{1}{2}$

26. An electric kettle is rated 230 V; 1000 W. Calculate the resistance of its heating element when in operation.

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26. An electric kettle is rated 230 V; 1000 W. Calculate the resistance of its heating element when in operation.

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2

$$P = 1000 \text{ W}, V = 230 \text{ V}; \text{ Formula } = P = \frac{V^2}{R}$$

$$R = \frac{V^2}{P}$$

$$= \frac{(230)^2}{1000}$$

$$= 52.9 \, \Omega$$

$\frac{1}{2}$

1

$\frac{1}{2}$

15. The resistance of a wire is $40\ \Omega$. If 2 A of current flows through this wire for 50 seconds, the heat produced in the wire is

(A) 4000 J

(B) 8000 J

(C) 16000 J

(D) 80000 J

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15. The resistance of a wire is $40\ \Omega$. If 2 A of current flows through this wire for 50 seconds, the heat produced in the wire is

(A) 4000 J

(B) 8000 J

(C) 16000 J

(D) 80000 J

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(B)/ 8000 J

36. (a) Define Electric power. Express electric power of an electric appliance in terms of its resistance (R) and the voltage / potential difference (V) at which it operates.
- (b) The rated voltage of an air-conditioner is 220 V. This air-conditioner consumes 22 units of electric energy in 10 hours, calculate :
- (i) Power rating of the air-conditioner,
 - (ii) Current drawn by the air-conditioner, and
 - (iii) Resistance of the air-conditioner

5

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(a)	• Electric Power: It is the rate at which electric energy is dissipated or consumed.	1
	• Expression: $P = V^2/R$	1
(b)	(i) $\text{Power} = \frac{\text{Electrical energy consumed}}{\text{Time}}$	$\frac{1}{2}$
	$= \frac{22 \text{ kWh}}{10 \text{ h}} = 2.2 \text{ kW or } 2200 \text{ W}$	$\frac{1}{2}$
	(ii) $\text{Current} = \frac{\text{Power}}{\text{Voltage}}$	$\frac{1}{2}$
	$= \frac{2200 \text{ W}}{220 \text{ V}} = 10 \text{ A}$	$\frac{1}{2}$
	(iii) $\text{Resistance} = \frac{(\text{Voltage})^2}{\text{Power}}$	$\frac{1}{2}$
	$= \frac{(220)^2}{2200} = 22 \text{ } \Omega$	$\frac{1}{2}$

(B) Calculate potential difference across a $4\ \Omega$ resistor that produces 100 J of heat every second.

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(B) Calculate potential difference across a $4\ \Omega$ resistor that produces 100 J of heat every second.

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(B) Heat, $H = 100\text{ J}$

$$R = 4\ \Omega$$

$$t = 1\text{ s}$$

$$H = \frac{V^2 t}{R}$$

$$\Rightarrow V = \sqrt{\frac{HR}{t}} = \sqrt{\frac{400}{1}} = 20\text{ V}$$

1

$\frac{1}{2} + \frac{1}{2}$

(B) Prove that if the current through a resistor is increased by 100%, then the increase in power dissipated through the resistor will be 300%.

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- (B) Prove that if the current through a resistor is increased by 100%, then the increase in power dissipated through the resistor will be 300%.

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$$(B) P_1 = I^2 R$$

$$P_2 = (2I)^2 R = 4I^2 R \text{ [100\% increase in current means current becomes } 2I]$$

$$\therefore \text{Increase in power dissipated} = P_2 - P_1 = 4I^2 R - I^2 R = 3I^2 R \\ = 3P_1$$

$$\text{Percentage increase in power dissipated} = \frac{3P_1}{P_1} \times 100 = 300\%$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

25. (A) State two applications of Joule's heating in domestic electric circuit.
- (B) (a) Establish the relationship between the commercial unit of electric energy and the SI unit of electric energy.

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25. (A) State two applications of Joule's heating in domestic electric circuit.

(B) (a) Establish the relationship between the commercial unit of electric energy and the SI unit of electric energy.

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(A) Electric bulb / electric iron / Electric fuse / Electric heater / electric Oven

(Any two)

(B) (a) $1 \text{ kWh} = 1000 \text{ watt} \times 3600 \text{ second}$
 $= 3.6 \times 10^6 \text{ watt second}$
 $= 3.6 \times 10^6 \text{ Joule (J)}$

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

18. *Assertion (A) :* The commercial unit of electric energy is kilowatt hour (kWh).

Reason (R) : $1 \text{ kWh} = 10^6 \text{ joule (J)}$

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18. *Assertion (A) :* The commercial unit of electric energy is kilowatt hour (kWh).

Reason (R) : $1 \text{ kWh} = 10^6 \text{ joule (J)}$

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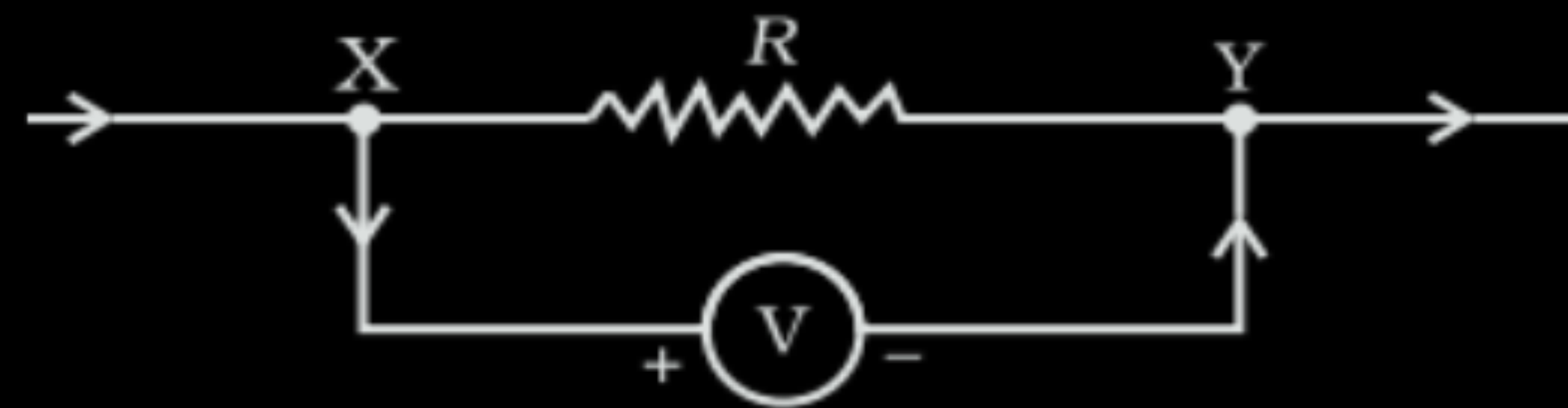
- (b) (i) What is meant by heating effect of electric current ? Derive an expression for the amount of heat produced in a conductor of resistance R through which a current I flows for a time t .

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- (ii) Calculate the amount of heat produced in a conductor of resistance $10\ \Omega$ through which a current of 2 A flows for 100 seconds.

(i)

- When electric current is passed through a resistor electric energy is dissipated and appears as heat energy.
- When charge Q flows through a conductor of resistance R having potential difference of V , the workdone W is given by-



$$\begin{aligned} W &= V \times Q = V \times I \times t \\ &= IR \times I \times t = I^2 R t \end{aligned}$$

($\because V = I \times R$ – Ohm's Law)

When circuit is purely resistive electric energy is converted into heat energy.

$$H = I^2 R t$$

(ii) Here $I = 2A$; $R = 10\Omega$; $t = 100s$

$$H = I^2 R t$$

$$= (2A)^2 \times 10\Omega \times 100s$$

16. The resistance of a resistor is increased four times its initial value. If other parameters (current and time) remain unchanged, the percentage change in the amount of heat produced in the resistor would be :

(A) 100%

(B) 200%

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(C) 300%

(D) 400%

16. The resistance of a resistor is increased four times its initial value. If other parameters (current and time) remain unchanged, the percentage change in the amount of heat produced in the resistor would be :

(A) 100%

(B) 200%

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(C) 300%

(D) 400%

(C)

- (b) (i) Obtain an expression for the amount of heat produced in a conductor of resistance R when a steady current I flows through it for a time t .
- (ii) An electrical heater of resistance $10\ \Omega$ draws a steady current of 5 A from the service mains for 2 hours. Find the power of this heater and the amount of energy consumed by the heater in kilowatt hour.

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(i) When Q amount of charge flows through a potential difference of V, Then the amount of work done, $W = V Q$ -----(1)	$\frac{1}{2}$
$P = W/t$ $= VQ/t$ [From (1)] $= V I$ [I = Q/t] -----(2)	$\frac{1}{2}$
Work done=heat produced, $H = P \times t$ $= VIt$ [From (2)] $= I^2Rt$ [From Ohm's law; $V = IR$]	$\frac{1}{2}$
(ii) Here, $R = 10 \Omega$, $I = 5 A$, $t = 2 h$, $P = ?$; $E = ?$ $P = I^2 R$ $= (5A)^2 \times 10 \Omega = 25 \times 10 W$ $= 250 W$	$\frac{1}{2}$
Energy / Work done = $P \times t$ $= 250 W \times 2 h = 500 Wh$ $= 0.5 kWh$	1
	$\frac{1}{2}$
	1

32. A 2000 W heater has a resistance of about $25\ \Omega$, whereas a 100 W bulb has a resistance of $500\ \Omega$. When 220 V is applied on these, then which of the two

- (a) can carry large currents ?
- (b) may be used with an electrical circuit having 1.0 A rating ?
- (c) will be fitted with a 15 A electric board and not with a 5 A electric board ?

Justify your answer in each case.

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(a) 2000 W heater , For heater, $I_1 = \frac{P}{V} = 9.09 \text{ A}$; For Bulb, $I_2 = \frac{P}{V} = 0.45 \text{ A}$ $I_1 > I_2$	$\frac{1}{2}$ $\frac{1}{2}$
(b) 100 W bulb $I_2 = \frac{P}{V} = 0.45 \text{ A}$ As it draws only 0.45 A which is less than 1 A.	$\frac{1}{2}$ $\frac{1}{2}$
(c) 2000 W heater $I_1 = \frac{P}{V} = 9.09 \text{ A}$ As the current drawn is 9.09 A which is higher than 5.0 A.	$\frac{1}{2}$ $\frac{1}{2}$

1. An electric iron of resistance $20\ \Omega$ draws a current of $5\ \text{A}$. The heat developed in the iron in 30 seconds is :

(A) $15000\ \text{J}$

(B) $6000\ \text{J}$

(C) $1500\ \text{J}$

(D) $3000\ \text{J}$

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1. An electric iron of resistance $20\ \Omega$ draws a current of $5\ \text{A}$. The heat developed in the iron in 30 seconds is :

(A) $15000\ \text{J}$

(B) $6000\ \text{J}$

(C) $1500\ \text{J}$

(D) $3000\ \text{J}$

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(A)/15000 J

26. State Joule's law of heating. How is this effect useful in electric circuits where fuse is used as a safety device ?

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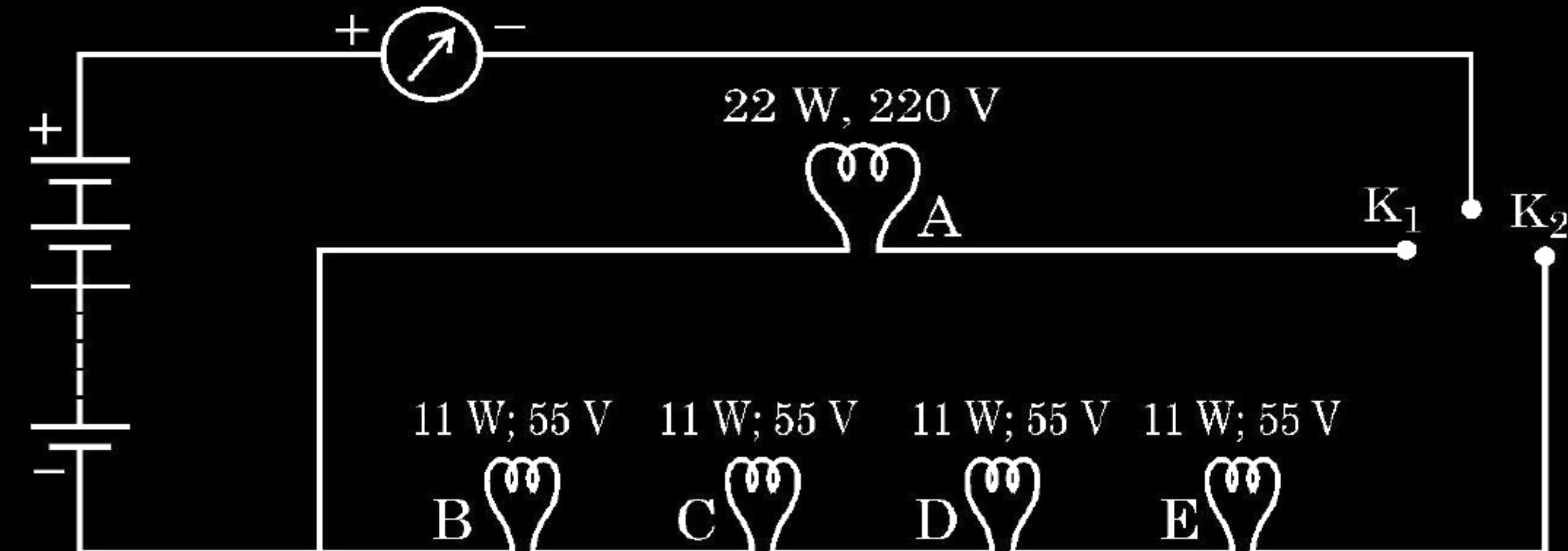
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26. State Joule's law of heating. How is this effect useful in electric circuits where fuse is used as a safety device ?

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2

- | | |
|---|-------------------|
| <ul style="list-style-type: none">• Joule's Law – Heat produced in a resistor is directly proportional to:<ul style="list-style-type: none">-Square of current for a given resistance-Resistance for a given conductor and-Time for which the current flows through the resistor,• If any unduly high electric current flows through the circuit, the temperature of the fuse wire increases. This melts the fuse wire and breaks the circuit. | <p>1</p> <p>1</p> |
|---|-------------------|



39. In a domestic circuit five LED bulbs are arranged as shown. The source voltage is 220 V and the power rating of each bulb is marked in the circuit diagram. Based on the following circuit diagram, answer the following questions :

- (a) State what happens when
- key K_1 is closed.
 - key K_2 is closed.
- (b) Find the current drawn by the bulb B when it glows.
- (c) Calculate
- the resistance of bulb B, and
 - total resistance of the combination of four bulbs B, C, D and E.

OR

- (c) What would happen to the glow of all the bulbs in the circuit when keys K_1 and K_2 both are closed and the bulb C suddenly get fused ?

Give reason to justify your answer.

(a)		
(i)	Bulb A glows	$\frac{1}{2}$
(ii)	Bulbs B, C, D and E glow	$\frac{1}{2}$
(b)	$P = V \times I$	$\frac{1}{2}$
	$11 = 55 \times I$	
	$I = \frac{1}{5} = 0.2 \text{ amp}$	$\frac{1}{2}$
(c)		
(i)	Resistance of bulb B, $R = \frac{V}{I} = \frac{55 \text{ V}}{0.2 \text{ A}} = 275 \Omega$	1
	(alternative formula for calculation $R = \frac{V^2}{P}$)	
(ii)	Total resistance of the series combination of four bulbs $= 4 \times 275 = 1100 \Omega$ OR	1
(c)		
•	Bulb A will keep glowing with same brightness.	$\frac{1}{2}$
•	Other bulbs i.e., B, D and E will stop glowing.	$\frac{1}{2}$
	Reason :	
	As the bulbs B, D and E are connected in series with fused bulb C, so no current flows through them and thus they will not glow. The bulb A remains unaffected as it is connected in parallel combination.	1

38. When electric current flows in a purely resistive circuit electrical energy gets fully converted into heat energy. The amount of heat produced (H) in the circuit is found to be directly proportional to (i) the square of current (I^2) (ii) the resistance (R) of the conductor and (iii) the time (t) for which current flows. In other words $H = I^2Rt$. Electrical devices such as an electric fuse, electric heater, electric iron etc. are all based on this effect called heating effect of electric current. 4
- (a) List two properties of heating elements. 1
- (b) List two properties of electric fuse. 1
- (c) Name the principle on which an electric fuse works. Explain how a fuse wire is capable of saving electrical appliances from getting damaged due to accidentally produced high currents. 2

OR

- (c) The power of an electric heater is 1100 W. If the potential difference between the two terminals of the heater is 220 V, find the current flowing in the circuit. What will happen to an electric fuse of rating 5 A connected in this circuit ?

• Higher resistivity than pure metals • Do not oxidise (burn) at high temperature.	1/2 , 1/2
(c) • Higher resistivity than pure metals • Low melting point.	1/2 , 1/2
(c) • Heating effect of electric current	1
• When high current flows in the circuit accidentally, the fuse wire melts and breaks the circuit .	1
OR	
(c) P = 1100 W; V = 220 V, I = ?	
P = VI	1/2
$I = \frac{P}{V} = \frac{1100\text{ W}}{220\text{ V}} = 5\text{A}$	1/2
• No effect on the fuse of 5A rating.	1

- 34. (A) (i)** State whether the currents and potential difference in all the bulbs will be same or different when in a circuit three bulbs of :
- (a) same wattage are connected in series.
 - (b) same wattage are connected in parallel.
 - (c) different wattage are connected in series.
 - (d) different wattage are connected in parallel.
- (ii)** Two identical resistors of $24\ \Omega$ each are connected to a battery of 6 V . Calculate the ratio of the power consumed by the resulting combinations with (a) minimum resistance and (b) maximum resistance.

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(A)(i)(a) same current and same potential difference. (b) same current and same potential difference (c) same current but different potential difference (d) different current but same potential difference.	$\frac{1}{2} \times 4$
(ii) (a) Minimum resistance – When resistors are in parallel $\frac{1}{R} = \frac{1}{24} + \frac{1}{24}$	$\frac{1}{2}$
$\therefore R_P = 12 \Omega$	$\frac{1}{2}$
Power consumed $P_1 = \frac{V^2}{R_P}$ $= \frac{6V \times 6V}{12 \Omega}$	$\frac{1}{2}$
$= 3W$	$\frac{1}{2}$
(b) $\therefore R_S = 24 \Omega + 24 \Omega = 48 \Omega$ Power consumed $P_2 = \frac{V^2}{R_S}$ $= \frac{6V \times 6V}{48 \Omega}$ $= \frac{3}{4}W$	$\frac{1}{2}$
$\therefore \frac{P_1}{P_2} = \frac{3}{\frac{3}{4}} = \frac{4}{1}$ $\Rightarrow P_p : P_s = 4 : 1$	$\frac{1}{2}$

(B) Draw a schematic diagram of a circuit consisting of a battery of six 2V cells, a $6\ \Omega$ resistor, a $12\ \Omega$ resistor and a $18\ \Omega$ resistor and a plug key all connected in series. Calculate the following (when key is closed) :

5

- (i) Electric current flowing in the circuit.
- (ii) Potential difference across $18\ \Omega$ resistor.
- (iii) Electric power consumed in $18\ \Omega$ resistor.

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(b)

$$\therefore R_s = 24 \Omega + 24 \Omega = 48 \Omega$$

$$\begin{aligned} \text{Power consumed } P_2 &= \frac{V^2}{R_s} \\ &= \frac{6 V \times 6 V}{48 \Omega} \\ &= \frac{3}{4} W \end{aligned}$$

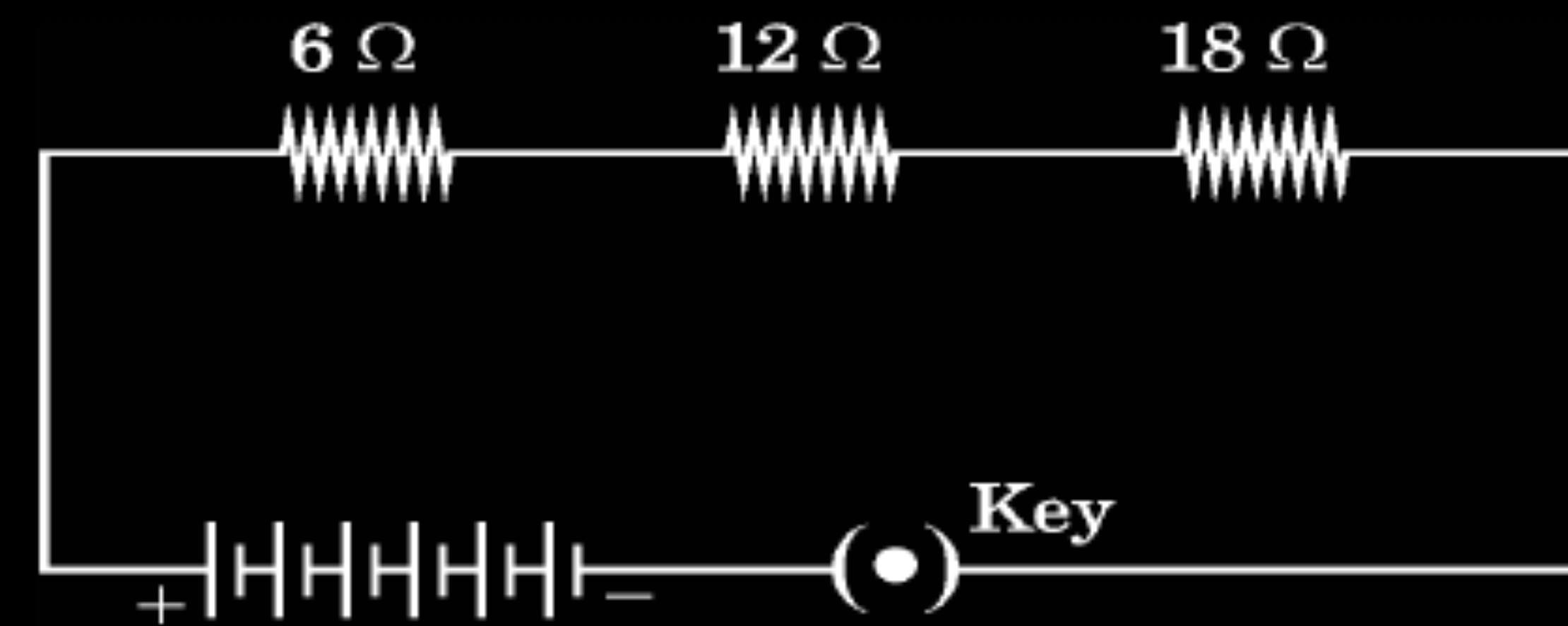
 $\frac{1}{2}$

$$\therefore \frac{P_1}{P_2} = \frac{\frac{3}{4}}{\frac{3}{4}} = \frac{4}{1} \quad \text{from } P_1 \text{ and } P_2$$

$$\Rightarrow P_p : P_s = 4 : 1$$

 $\frac{1}{2}$ **OR**

(B)



12 V (Six cells of 2 V each)

2

$$(i) \text{Current} = \frac{V}{R} = \frac{12}{(6+12+18) \Omega} = \frac{1}{3} A$$

1

$$(ii) \text{Potential difference across } 18 \Omega \text{ resistor} = I \times R = \frac{1}{3} A \times 18 \Omega = 6 V$$

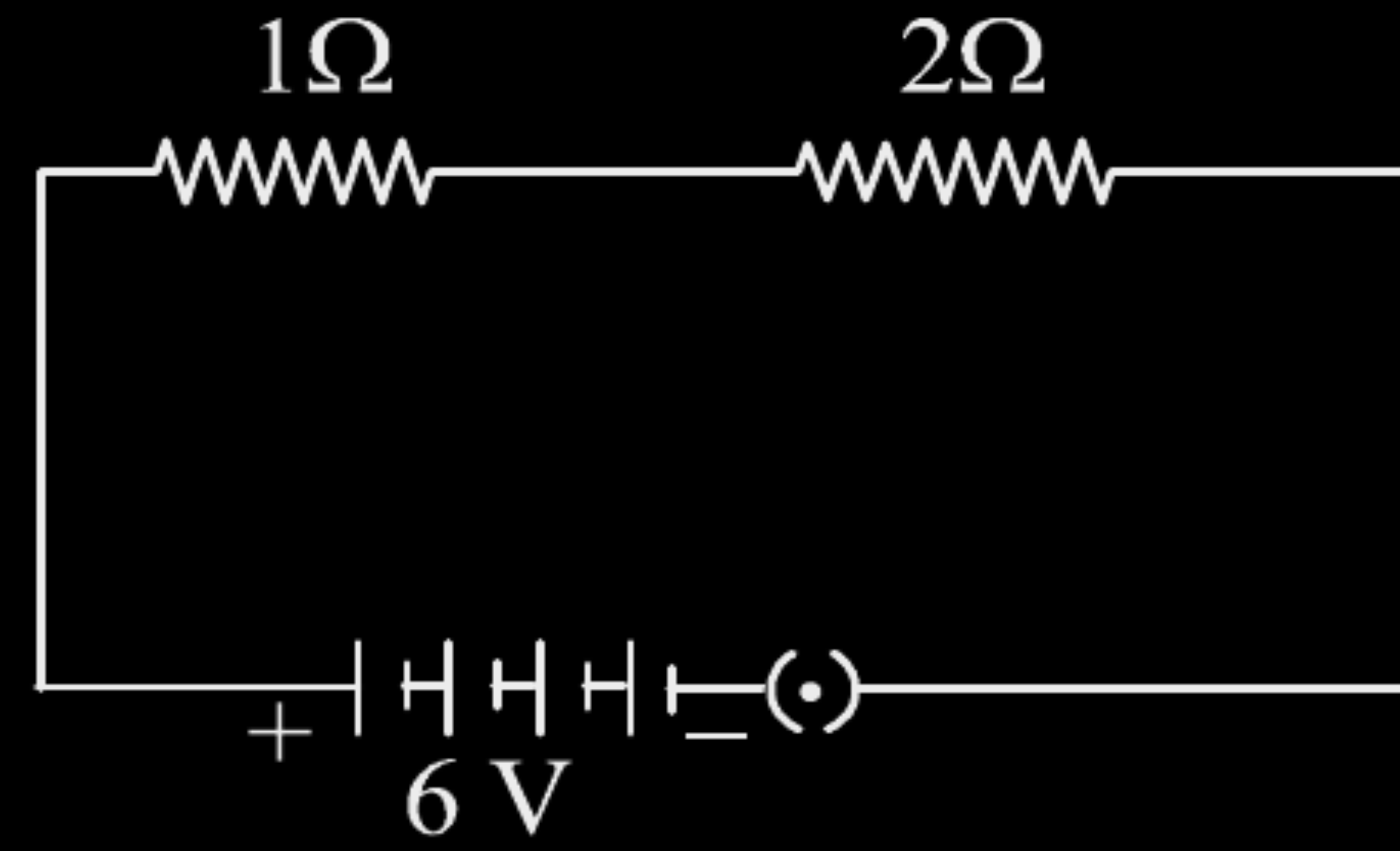
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$$(iii) \text{Power consumed in } 18 \Omega \text{ resistor} = V \times I = 6 V \times \frac{1}{3} A = 2 W$$

1

(B) In the given circuit calculate the power consumed in watts in the resistor of $2\ \Omega$:

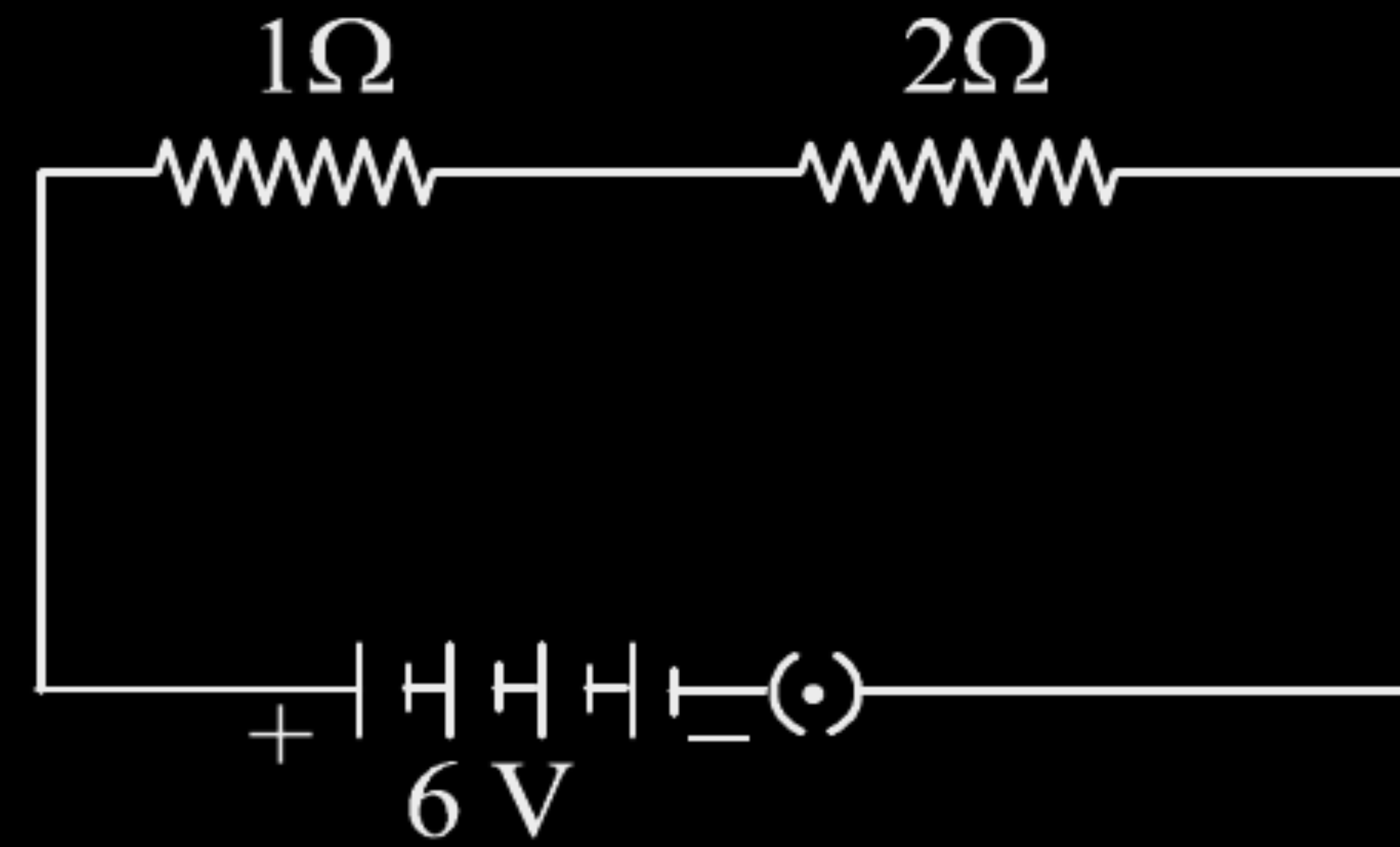
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(B) In the given circuit calculate the power consumed in watts in the resistor of $2\ \Omega$:

2



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(B) Equivalent resistance = $R_1 + R_2 = 1\ \Omega + 2\ \Omega = 3\ \Omega$

$\frac{1}{2}$

$$I = \frac{V}{R}$$

$$= \frac{6\text{ V}}{1\ \Omega + 2\ \Omega} = \frac{6\text{ V}}{3\ \Omega} = 2\text{ A}$$

$\frac{1}{2}$

Electric power, $P = I^2 R$

$\frac{1}{2}$

$$= (2\text{ A})^2 \times 2\ \Omega = 4 \times 2\text{ W} = 8\text{ W}$$

$\frac{1}{2}$

35. (A) (i) Define electric power. Express it in terms of potential difference (V) and resistance (R). **5**

(ii) An electric oven is designed to work on the mains voltage of 220 V. This oven consumes 11 units of electrical energy in 5 hours. Calculate :

- (a) power rating of the oven
- (b) current drawn by the oven
- (c) resistance of the oven when it is red hot

CBSE 2024

(A) (i)

- Electric power : Rate at which electrical energy is dissipated or consumed / Rate of supplying energy to maintain the flow of current through a circuit.

1

- $P = \frac{V^2}{R}$

1

(ii) (a) (1 unit = 1kWh)

$$\text{Power, } P = \frac{\text{Electrical energy consumed}}{\text{Time}}$$

$\frac{1}{2}$

$$= \frac{11\text{kWh}}{5\text{h}} = 2.2\text{kW or } 2200 \text{ W}$$

$\frac{1}{2}$

(b) $I = \frac{P}{V}$

$\frac{1}{2}$

$$= \frac{2200}{220} = 10\text{A}$$

$\frac{1}{2}$

(c) $R = \frac{V^2}{P}$

$\frac{1}{2}$

$$= \frac{(220)^2}{2200} = 22 \Omega$$

$\frac{1}{2}$

(Alternate formula can be used)

- 36.** (a) The commercial unit of electric energy is commonly known as 'unit'. Write its actual name and establish the relationship between this unit and the SI unit of energy.
- (b) The current rating of a circuit is 1·0 A. How many LED lamps of power rating 8 W; 220 V can safely be used simultaneously in this circuit ?

CBSE 2023 C

- 36.** (a) The commercial unit of electric energy is commonly known as ‘unit’. Write its actual name and establish the relationship between this unit and the SI unit of energy.
- (b) The current rating of a circuit is 1.0 A. How many LED lamps of power rating 8 W; 220 V can safely be used simultaneously in this circuit ?

CBSE 2023 C

5

(a)

- kilowatt hour or kWh
- $1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ seconds}$
 $= 3.6 \times 10^6 \text{ watt seconds}$
 $1 \text{ kWh} = 3.6 \times 10^6 \text{ Joule}$

1

$\frac{1}{2}$

1

(b) $P = 8 \text{ W}$; $V = 220 \text{ V}$; Current rating = 1.0 A

$$\text{Current through each lamp} = \frac{8 \text{ W}}{220 \text{ V}}$$

1

$$\begin{aligned} \text{Number of lamp} &= \frac{\text{current rating}}{\text{current through each lamp}} \\ &= \frac{1.0 \text{ A}}{\frac{8 \text{ W}}{220 \text{ V}}} = \frac{220}{8} = 27.75 \end{aligned}$$

$\frac{1}{2}$

So, 27 lamps can safely be used in the circuit.

1

14. Two LED bulbs of 12 W and 6 W are connected in series. If the current flowing through 6 W bulbs is 0.005 A, the current that flows through 12 W bulb is :

(a) 0.005 A

(b) 0.01 A

(c) 0.0025 A

(d) 0.02 A

CBSE 2023 C

14. Two LED bulbs of 12 W and 6 W are connected in series. If the current flowing through 6 W bulbs is 0.005 A, the current that flows through 12 W bulb is :

(a) 0.005 A

(b) 0.01 A

(c) 0.0025 A

(d) 0.02 A

CBSE 2023 C

- 36.** (a) Derive an expression for the amount of heat produced when a current flows through a resistor for a given time. Use this expression to state Joule's law of heating.
- (b) 200 joules of heat energy is produced each second in a resistor of $8\ \Omega$. Find :
- (i) The current flowing through the resistor, and
 - (ii) The potential difference across the ends of the resistor.

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(a)		
Consider a current I flowing through a resistor of resistance R having a potential difference V across its ends.		
Let t be the time during which charge Q flows across		
$\therefore$ Work done in moving charge $W = VQ$		
$\therefore$ Power input to the circuit by the source $P = \frac{VQ}{t} = VI$		
$\therefore$ Energy supplied by the source to the circuit in time $t = VIt$	$\frac{1}{2}$	
$\hspace{15em} = IR \cdot It$	$\frac{1}{2}$	
This energy is dissipated in the form of heat if the circuit is purely resistive.		
$\therefore H = I^2Rt$	$\frac{1}{2}$	
This is the required expression for Joule's law of heating which implies that the heat produced H in a resistor is		
(i) directly proportional to square of current (I^2) for given R		$\frac{1}{2}$
(ii) directly proportional to resistance (R) for given I		$\frac{1}{2}$
(iii) directly proportional to time t for which the current flows through the resistor		$\frac{1}{2}$
(b) $H = 200 \text{ J}$, $t = 1 \text{ s}$, $R = 8 \Omega$		
(i) Using the formula $H = I^2Rt$, we have $I^2 = \frac{H}{Rt}$		
$\hspace{15em} = \frac{200 \text{ J}}{8 \times 1 \Omega \text{ t}} =$		
$\hspace{15em} = 25$		
$\hspace{15em} \Rightarrow I = 5\text{A}$		1
(ii) $V = IR$		$\frac{1}{2}$
$\hspace{15em} = 5\text{A} \times 8 \Omega = 40\text{V}$		$\frac{1}{2}$

(b) In an electric circuit three bulbs of 100 W each are connected in series to a source. In another circuit set of three bulbs of the same wattage are connected in parallel to the same source. 5

- (i) Will the bulb in the two circuits glow with the same brightness ? Justify your answer.
- (ii) Now, let one bulb in both the circuits get fused. Will the rest of the bulbs continue to glow in each circuit ? Give reason for your answer.

CBSE 2023

- (b) In an electric circuit three bulbs of 100 W each are connected in series to a source. In another circuit set of three bulbs of the same wattage are connected in parallel to the same source. 5
- (i) Will the bulb in the two circuits glow with the same brightness ? Justify your answer.
- (ii) Now, let one bulb in both the circuits get fused. Will the rest of the bulbs continue to glow in each circuit ? Give reason for your answer.

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(b) (i) No

- In series combination overall resistance will increase hence decreasing the current . Potential difference also divides. Therefore power consumption is less by each bulb and glows with less brightness.
- In parallel combination each bulb will get the required potential difference hence the required current and will glow with its normal brightness.

(ii) None of the bulb glows in series combination as the circuit gets broken and current stops flowing.

In parallel combination the other two bulbs will glow with same brightness as the same voltage is available to them.

1

1

1

1

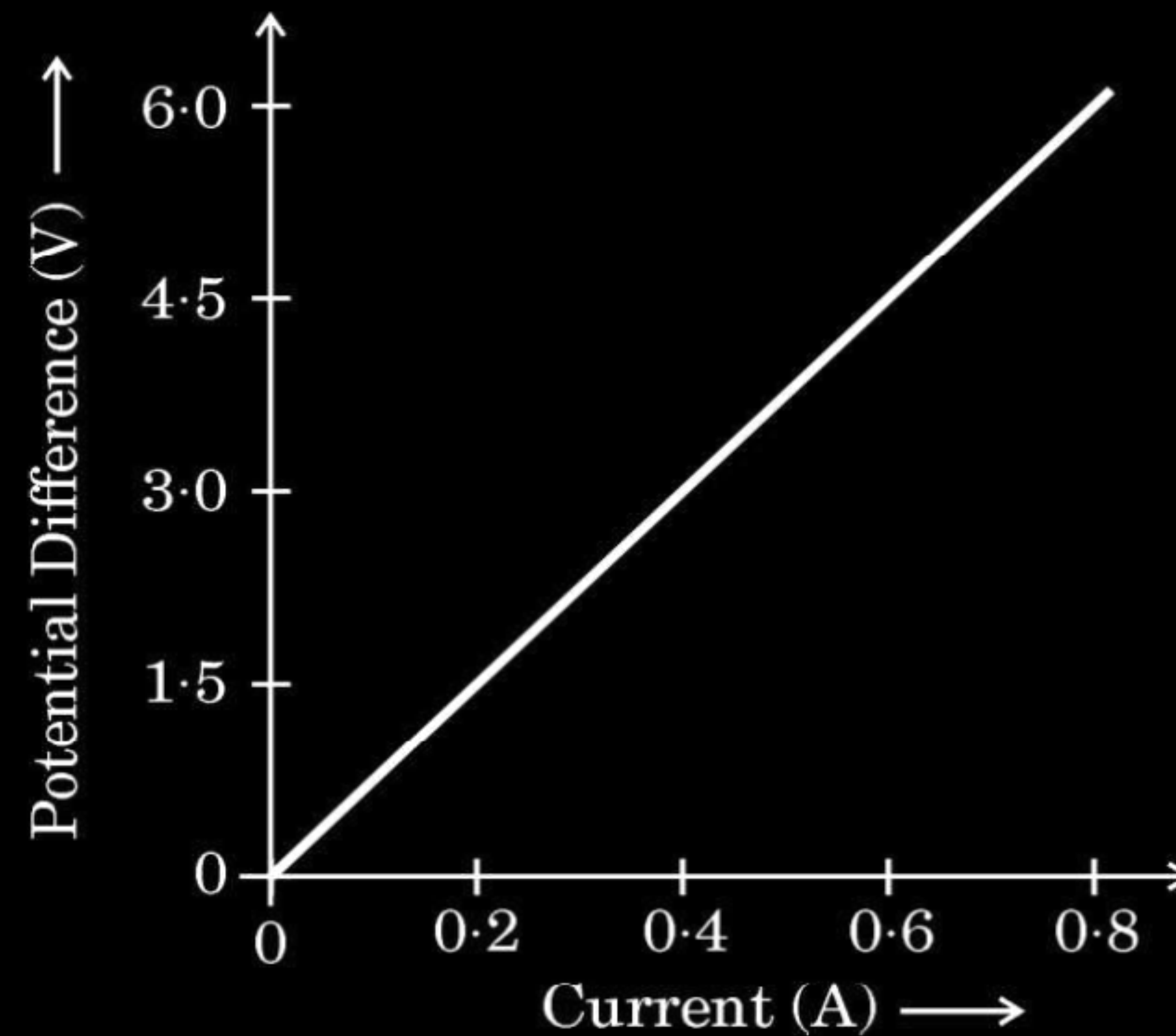
1

- 35.** (a) Define electric power and state its SI unit. The commercial unit of electrical energy is known as ‘unit’. Write the relation between this ‘unit’ and joule.
- (b) In a house, 2 bulbs of 50 W each are used for 6 hours daily and an electric geyser of 1 kW is used for 1 hour daily. Calculate the total energy consumed in a month of 30 days and its cost at the rate of ₹ 8.00 per kWh.

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(a) • The rate at which electrical energy is dissipated / consumed in a circuit.	1
• Watt or W	$\frac{1}{2}$
• 1 unit (1 kWh) = 3.6×10^6 joule (J)	$\frac{1}{2}$
(b) $E = P \times T$	$\frac{1}{2}$
Electrical energy, E_1 , consumed by 2 bulbs of 50 W each is	$\frac{1}{2}$
$E_1 = 50 \times 2 \times 6 = 600 \text{ Wh}$	
Electrical energy, E_2 , consumed by electric geyser of 1 kW is	
$E_2 = 1000 \times 1 = 1000 \text{ Wh}$	$\frac{1}{2}$
Total energy consumed in the month of 30 days	
$[E_1 + E_2] \times 30 = [600 + 1000] \times 30 \text{ Wh}$	
$= 1600 \times 30 = 48000 \text{ Wh}$	
$= 48 \text{ kWh}$	$\frac{1}{2}$
Cost of using these electrical devices at the rate of ₹ 8.00 per kwh is	
₹ $48 \times 8 = ₹ 384$	1

36. (a) State Ohm's Law.
- (b) Name and define the physical quantity determined by the slope of $V - I$ curve given in the diagram. Use this graph to find the value of this physical quantity in SI units.



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- (c) Establish the relationship between 1 kWh and 1 joule.

(a) The potential difference V across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it provided its temperature remains the same. / $V \propto I$ or $V = IR$ (temperature remaining constant)	1
(b) Resistance Resistance is the property of a conductor to resist the flow of charges. $R = \text{slope} = \frac{y_2 - y_1}{x_2 - x_1} / \frac{V}{I} = \frac{6 - 0}{0.8 - 0}$ $= \frac{60}{8} = 7.5 \, \Omega$	1 1 1/2 1/2
(c) $1 \text{ kWh} = 1000 \frac{J}{\text{sec}} \times 3600 \text{ sec}$ $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$	1

14. Two LED bulbs of 12W and 6W are connected in series. If the current through 12W bulb is 0.06A the current through 6W bulb will be :

1

(a) 0.04A

(b) 0.06A

(c) 0.08A

(d) 0.12A

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14. Two LED bulbs of 12W and 6W are connected in series. If the current through 12W bulb is 0.06A the current through 6W bulb will be :

1

(a) 0.04A

(b) 0.06A

(c) 0.08A

(d) 0.12A

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(b)

16. The resistance of a resistor is reduced to half of its initial value. If other parameters of the electrical circuit remain unaltered, the amount of heat produced in the resistor will become :

1

- (a) four times
- (b) two times
- (c) half
- (d) one fourth

CBSE 2023

16. The resistance of a resistor is reduced to half of its initial value. If other parameters of the electrical circuit remain unaltered, the amount of heat produced in the resistor will become :

1

- (a) four times
- (b) two times
- (c) half
- (d) one fourth

CBSE 2023

(b)

16. Two LED bulbs of 10W and 5W are connected in series. If the current flowing through 5W bulb is 0.005A, the current flowing through 10W bulb is :

- (a) 0.02A
- (b) 0.01A
- (c) 0.005A
- (d) 0.0025A

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16. Two LED bulbs of 10W and 5W are connected in series. If the current flowing through 5W bulb is 0.005A, the current flowing through 10W bulb is :

- (a) 0.02A
- (b) 0.01A
- (c) 0.005A
- (d) 0.0025A

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16. In a resistive circuit if the current is increased to two times, the percentage change in the amount of heat dissipated in the circuit would be :

(a) 400%

(b) 300%

(c) 200%

(d) 100%

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(b)

16. If the current I through a resistor is increased by 100% (at constant temperature), the increase in power dissipated will be 1

(a) 100%

(b) 200%

(c) 300%

(d) 400%

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16. If the current I through a resistor is increased by 100% (at constant temperature), the increase in power dissipated will be 1

(a) 100%

(b) 200%

(c) 300%

(d) 400%

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(c)

35. (a) An electric iron consumes energy at a rate of 880 W when heating is at the maximum rate and 330 W when the heating is at the minimum. If the source voltage is 220 V, calculate the current and resistance in each case.
- (b) What is heating effect of electric current ?
- (c) Find an expression for the amount of heat produced when a current passes through a resistor for some time.

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(a) When heating is at maximum rate.

Power, $P = 880 \text{ W}$

Voltage, $V = 220 \text{ V}$

$$\text{Current, } I = \frac{P}{V} = \frac{880}{220} = 4 \text{ A}$$

$$\text{Resistance, } R = \frac{V}{I} = \frac{220}{4} = 55 \, \Omega$$

When heating is at minimum rate

Power, $P = 330 \text{ W}$

Voltage, $V = 220 \text{ V}$

$$\text{Current, } I = \frac{P}{V} = \frac{330}{220} = \frac{3}{2} = 1.5 \text{ A}$$

$$\text{Resistance, } R = \frac{V}{I} = \frac{220}{1.5} = 146.6 \, \Omega$$

(b) When electric current is passed through a resistor, electrical energy is dissipated and appears as heat energy.

$$(c) \quad H = I^2 R t / H = V I t$$

12. For a heater rated 4.4 kW; 220 V, calculate the following : 3

(a) The current

(b) The resistance of the heater

(c) The energy consumed in kWh in 5 hours

CBSE 2022 C

12. For a heater rated 4.4 kW; 220 V, calculate the following : 3

(a) The current

(b) The resistance of the heater

(c) The energy consumed in kWh in 5 hours

CBSE 2022 C

(a) $P = 4.4\text{kW}$ $V = 220\text{ V}$

$$P = VI$$

$$I = \frac{4.4 \times 1000\text{ W}}{220\text{ V}}$$

$$I = 20\text{ A}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

(b) $P = \frac{V^2}{R}$

$$R = \frac{V^2}{P}$$

$$R = \frac{220\text{ V} \times 220\text{ V}}{4.4 \times 10^3\text{ W}}$$

$$R = 11\ \Omega$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

(c) $E = P \times t$

$$E = 4.4\text{kW} \times 5\text{ hours}$$

$$= 22.0\text{ kWh}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- 13.** Define the term electric power. An electric device of resistance R when connected across an electric source of voltage V draws a current I . Derive an expression for the power in terms of resistance R and voltage V . What is the power of a device of resistance $400\ \Omega$ operating at 200 V ?

3

CBSE 2022 C

13. Define the term electric power. An electric device of resistance R when connected across an electric source of voltage V draws a current I . Derive an expression for the power in terms of resistance R and voltage V . What is the power of a device of resistance $400\ \Omega$ operating at 200 V ?

3

CBSE 2022 C

- Rate at which electrical energy is dissipated / consumed in an electric circuit.
- Power $P = VI$

$$= \frac{V^2}{R} \quad (I = V/R)$$
- $$P = \frac{V^2}{R} = \frac{(200\text{ V})^2}{400\ \Omega} = 100\ \Omega$$

1

$\frac{1}{2}$

$\frac{1}{2}$

1

12. Give reason for the following :

- (a) Tungsten is used exclusively for the filaments of electric lamps.
- (b) Heating elements of electric heating devices are made of alloys rather than a pure metal.
- (c) Series arrangement is not used for domestic electric circuits.

CBSE 2022

12. Give reason for the following :

- (a) Tungsten is used exclusively for the filaments of electric lamps.
- (b) Heating elements of electric heating devices are made of alloys rather than a pure metal.
- (c) Series arrangement is not used for domestic electric circuits.

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(a) Tungsten has a high melting point. So it does not melt when electric current flows through it.	1
(b) Alloys used for heating element have generally high melting point / high resistivity / Do not oxidise even at high temperature.	1
(c) If one appliance is switched off (or damaged) all other appliances stop working. / Different appliances require different current which is not available in series arrangement. (or any other)	1

9. An electric motor rated 1100 W is connected to 220 V mains. Find :

3

- (i) The current drawn from the mains,
- (ii) Electric energy consumed if the motor is used for 5 hours daily for 6 days.
- (iii) Total cost of energy consumed if the rate of one unit is ₹ 5.

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9. An electric motor rated 1100 W is connected to 220 V mains. Find :

3

- (i) The current drawn from the mains,
- (ii) Electric energy consumed if the motor is used for 5 hours daily for 6 days.
- (iii) Total cost of energy consumed if the rate of one unit is ₹ 5.

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Power (P) = 1100 W, V = 220 V

(i) Current drawn $= I = \frac{P}{V}$
 $= \frac{1100 \text{ W}}{220 \text{ V}} = 5 \text{ A}$

1/2

1/2

(ii) $E = P \times t$
 $= 1100 \text{ W} \times 5 \text{ h} \times 6 = 33000 \text{ Wh}$

1/2

1/2

(iii) Cost of one commercial unit = ₹ 5
Energy consumed = 33 kWh = 33 unit = $118.8 \times 10^6 \text{ J}$
Cost of 33 unit = $33 \times 5 = ₹ 165$

1/2

1/2

1. (a) An electric heater rated 1100 W operates at 220 V. Calculate
(i) its resistance, and (ii) the current drawn by it.

CBSE 2022

2

1. (a) An electric heater rated 1100 W operates at 220 V. Calculate
(i) its resistance, and (ii) the current drawn by it.

CBSE 2022

2

(a) $P(\text{Power}) = V(\text{Potential difference}) \times I(\text{Current})$

Here $P = 1100 \text{ W}$, $V = 220 \text{ V}$, $I = ?$, $R = ?$

$$P = \frac{V^2}{R}$$

$\frac{1}{2}$

$$(i) R = \frac{V^2}{P}$$

$$= \frac{220^2 \text{ V} \times 220 \text{ V}}{1100 \text{ W}}$$

$$= 44 \Omega$$

$\frac{1}{2}$

$$(ii) I = \frac{V}{R}$$

$\frac{1}{2}$

$$I(\text{Current}) = \frac{V}{R} = \frac{220 \text{ V}}{44 \Omega} = 5 \text{ A}$$

$\frac{1}{2}$

- (a) What is the meaning of electric power of an electrical device ? Write its SI unit. **1½**
- (b) An electric kettle of 2kW is used for 2h. Calculate the energy consumed in
- (i) kilowatt hour and
 - (ii) joules. **1½**

CBSE 2022

- (a) What is the meaning of electric power of an electrical device ? Write its SI unit. 1½
- (b) An electric kettle of 2kW is used for 2h. Calculate the energy consumed in
- kilowatt hour and
 - joules. 1½

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(a)	Rate at which electric energy is dissipated / consumed in an electric circuit SI unit : watt / joule per second / volt . ampere	1 ½
(b)	$E = P \times t$	½
(i)	2 kW × 2h = 4kWh	½
(ii)	$4 \times 3.6 \times 10^6 \text{ joules} = 14.4 \times 10^6 \text{ J} / 1.44 \times 10^7 \text{ J}$	½

10. For a heater, rated 4 kW and 220 V, calculate the following :

3

(a) The current

(b) Energy consumed in 2 hours

(c) If 1 kWh is priced at ₹ 4.50, then the cost of energy consumed

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10. For a heater, rated 4 kW and 220 V, calculate the following :

3

(a) The current

(b) Energy consumed in 2 hours

(c) If 1 kWh is priced at ₹ 4.50, then the cost of energy consumed

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Given : $P = 4 \text{ kW} = 4000 \text{ W}$

$V = 220 \text{ V}$

(a)

$$P = VI$$

$$I = \frac{P}{V} = \frac{4000 \text{ W}}{220 \text{ V}} = 18.18 \text{ A}$$

(b)

$$E = P \times t$$

$$= 4 \text{ kW} \times 2 \text{ h} = 8 \text{ kWh}$$

(c) Cost of 1 kWh = ₹ 4.50

$$\text{Then } 8 \text{ kWh} = 8 \times 4.50 = ₹ 36.0$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

13. (a) (i) Define Electric Power and write its SI unit.

$$\frac{1}{2} + \frac{1}{2}$$

(ii) Two bulbs rated 100 W; 220 V and 60 W; 220 V are connected in parallel to an electric mains of 220 V. Find the current drawn by the bulbs from the mains.

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2

13. (a) (i) Define Electric Power and write its SI unit.

$$\frac{1}{2} + \frac{1}{2}$$

(ii) Two bulbs rated 100 W; 220 V and 60 W; 220 V are connected in parallel to an electric mains of 220 V. Find the current drawn by the bulbs from the mains.

CBSE 2022

2

(a) (i)

The rate at which electric energy is dissipated or consumed in an electric circuit.

S.I. unit—watt / V.A / joule per second

$\frac{1}{2}$

$\frac{1}{2}$

(a) (ii) • Current drawn by first bulb

$$I_1 = \frac{100 \text{ W}}{220 \text{ V}} = \frac{100}{220} \text{ ampere}$$

$\frac{1}{2}$

• Current drawn by second bulb

$$I_2 = \frac{60 \text{ W}}{220 \text{ V}} = \frac{60}{220} \text{ ampere}$$

$\frac{1}{2}$

$\frac{1}{2}$

Both the bulbs are in parallel

$$\text{Total current, } I = I_1 + I_2$$

$\frac{1}{2}$

$$= \left(\frac{100}{220} + \frac{60}{220} \right) \text{ ampere} = \frac{160}{220} \text{ A} = 0.73 \text{ A}$$

(Accept any other method)

(b) (i) State Joule's law of heating. Express it mathematically when an appliance of resistance R is connected to a source of voltage V and the current I flows through the appliance for a time t .

(ii) A $5\ \Omega$ resistor is connected across a battery of 6 volts.

Calculate the energy that dissipates as heat in 10 s.

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3

(b) i) This law states that heat produced in a resistor is—

- directly proportional to the square of current for a given resistance / $(H \propto I^2)$
- directly proportional to the resistance for a given current / $(H \propto R)$
- directly proportional to the time for which the current flows through the resistor / $(H \propto t)$
- $H = V I t$

ii)

$$V = 6\ \text{V}; R = 5\ \Omega; t = 10\ \text{s}$$

Energy dissipated as heat in $t = 10\ \text{s}$ is

$$H = \frac{V^2}{R} t$$

$$= \frac{(6\ \text{V})^2}{5\ \Omega} \times 10\ \text{s}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 29.** (a) Two lamps rated 100 W, 220 V and 10 W, 220 V are connected in parallel to 220 V supply. Calculate the total current through the circuit.
- (b) Two resistors X and Y of resistances $2\ \Omega$ and $3\ \Omega$ respectively are first joined in parallel and then in series. In each case the voltage supplied is 5 V.
- (i) Draw circuit diagrams to show the combination of resistors in each case.
- (ii) Calculate the voltage across the $3\ \Omega$ resistor in the series combination of resistors.

$$(a) I_1 = \frac{P_1}{V}$$

$$I_1 = \frac{100 \text{ W}}{220 \text{ V}} = \frac{10}{22} \text{ A}$$

$$I_2 = \frac{P_2}{V} = \frac{10}{220} = \frac{1}{22} \text{ A}$$

$$I = I_1 + I_2$$

$$= \left(\frac{10}{22} + \frac{1}{22} \right) \text{ A} = \frac{11}{22} \text{ A} = 0.5 \text{ A}$$

$$(ii) \text{ Net } R = R_1 + R_2 = 2 + 3 = 5 \Omega$$

$$I = \frac{V}{R_{net}} = \frac{5}{5} = 1 \text{ A}$$

∴ Voltage across 3Ω resistor :

$$\therefore V = 1 \times 3 = 3 \text{ V}$$

1/2

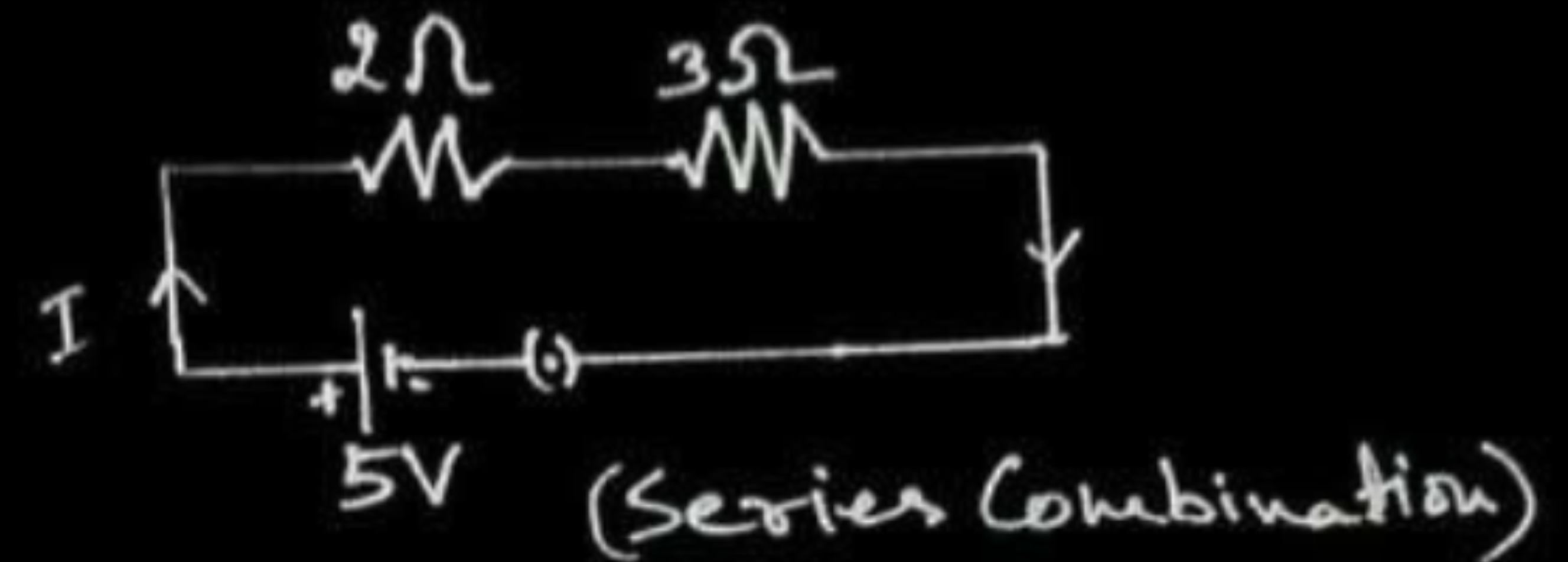
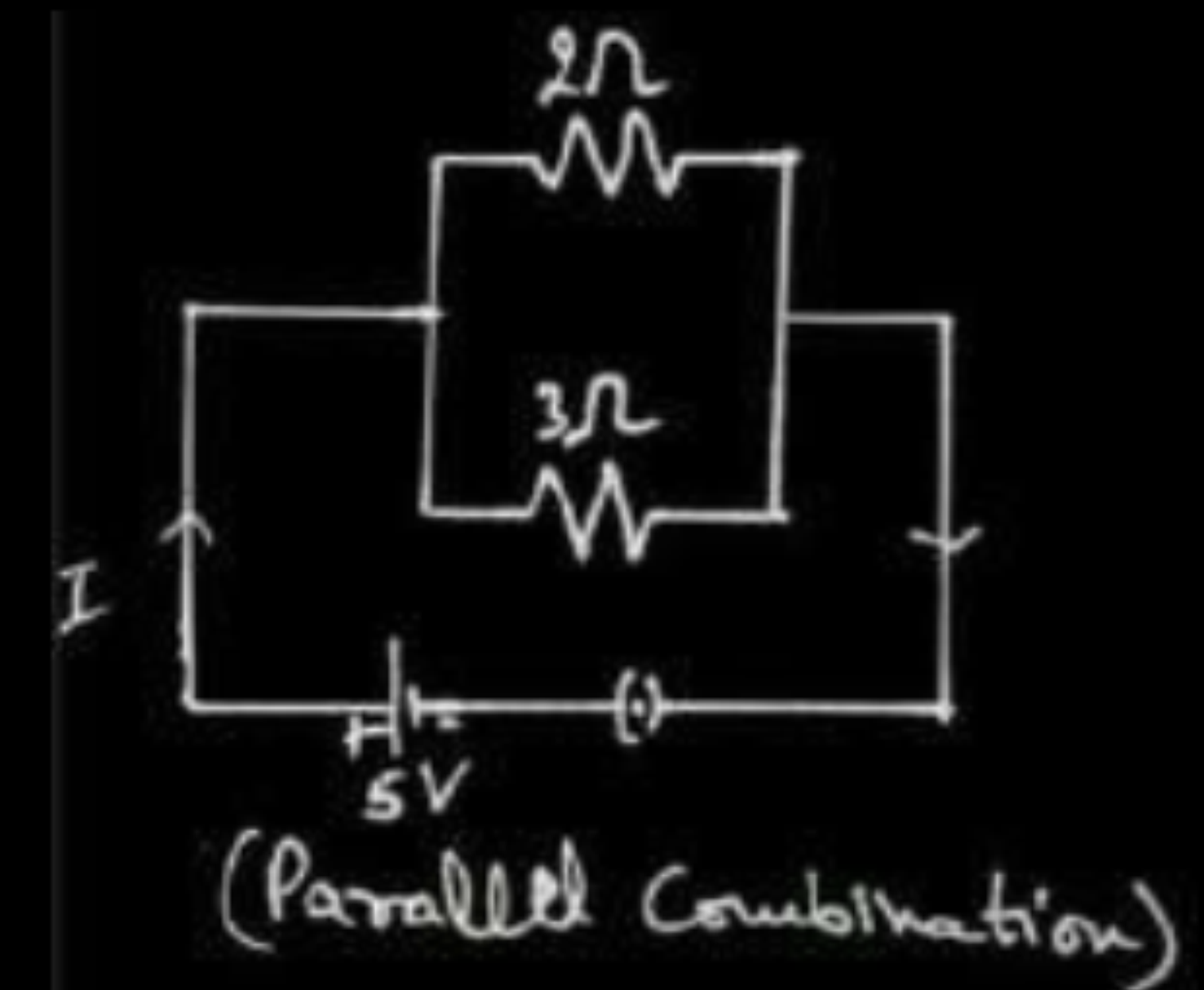
1/2

1/2

1

1

1/2



The resistance of a resistor is reduced to half of its initial value. In doing so, if other parameters of the circuit remain unchanged, the heating effects in the resistor will become

1

- (A) two times.
- (B) half.
- (C) one-fourth.
- (D) four times.

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The resistance of a resistor is reduced to half of its initial value. In doing so, if other parameters of the circuit remain unchanged, the heating effects in the resistor will become

1

- (A) two times.
- (B) half.
- (C) one-fourth.
- (D) four times.

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(B) /half

Two identical resistors, each of resistance $15\ \Omega$, are connected in (i) series, and (ii) parallel, in turn to a battery of $6\ \text{V}$. Calculate the ratio of the power consumed in the combination of resistors in each case.

5

CBSE 2020

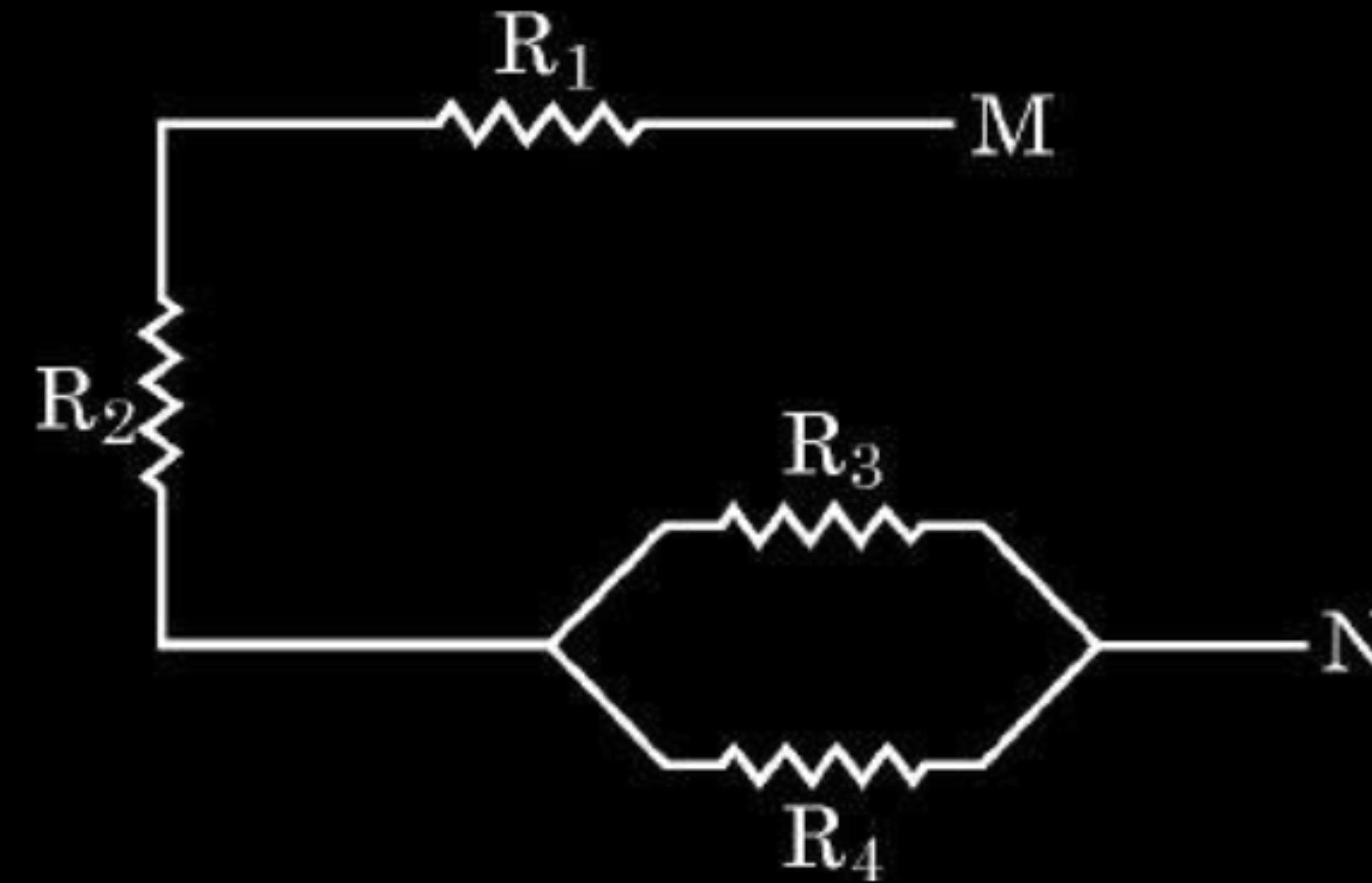
Two identical resistors, each of resistance $15\ \Omega$, are connected in (i) series, and (ii) parallel, in turn to a battery of 6 V . Calculate the ratio of the power consumed in the combination of resistors in each case.

5

$R_1 = R_2 = 15\Omega \quad V = 6V$	
i) In series :	
$R_s = R_1 + R_2 = 15\Omega + 15\Omega = 30\Omega$	$\frac{1}{2}$
$I = \frac{V}{R_s} = \frac{6V}{30\Omega} = 0.2\text{ A}$	$\frac{1}{2}$
$\therefore P_1 = VI = 6V \times 0.2\text{ A} = 1.2\text{ W}$	1
ii) In parallel	
$R_p = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{15 \times 15}{15 + 15} = \frac{225}{30} = 7.5\Omega$	$\frac{1}{2}$
$I = \frac{V}{R_p} = \frac{6V}{7.5\Omega} = 0.8\text{ A}$	$\frac{1}{2}$
$P_2 = VI = 6V \times 0.8\text{ A} = 4.8W$	1
Ratio of power = $\frac{P_1}{P_2} = \frac{1.2W}{4.8W} = \frac{1}{4}$	$\frac{1}{2}$
$\therefore P_1 : P_2 = 1 : 4$	$\frac{1}{2}$

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29. (a) For the combination of resistors shown in the following figure, find the equivalent resistance between M & N.



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- (b) State Joule's law of heating.
(c) Why we need a 5 A fuse for an electric iron which consumes 1 kW power at 220 V ?
(d) Why is it impracticable to connect an electric bulb and an electric heater in series ?

(a) R_3 and R_4 are in parallel combination .

$\therefore R_{\text{parallel}}$ is given by

$$\frac{1}{R_p} = \frac{1}{R_3} + \frac{1}{R_4}$$

$$\frac{1}{R_p} = \frac{R_4 + R_3}{R_3 R_4}$$

$$R_p = \frac{R_3 R_4}{R_4 + R_3}$$

Now, R_1 R_2 and R_p are in series.

$$\therefore R_{\text{eq}} = R_1 + R_2 + R_p$$

$$= R_1 + R_2 + \frac{R_3 R_4}{R_4 + R_3}$$

(b) The heat produced in a resistor is directly proportional to

- square of current for a given resistance.
- the resistance for a given current and
- the time for which the current flows through the resistor.

(Note : if a candidate writes $H = I^2 R t$ give full credit).

$$(c) P = V I \quad \text{or} \quad I = \frac{P}{V}$$

$$I = \frac{1000 \text{ watt}}{220 \text{ volt}} = 4.54 \text{ A}$$

Since 4.54 ampere current flows in the circuit , a 5 A fuse must be used.

(d) Electric bulb & electric heater will not get currents and voltages as per their requirement.

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Two bulbs of 100 W and 40 W are connected in series. The current through the 100 W bulb is 1 A. The current through the 40 W bulb will be :

(a) 0.4 A

(b) 0.6 A

(c) 0.8 A

(d) 1 A

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(d) / 1A

24. (a) Write the mathematical expression for Joule's law of heating

(b) Compute the heat generated while transferring 96000 coulomb of charge in two hours through a potential difference of 40 V.

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- (b) Compute the heat generated while transferring 96000 coulomb of charge in two hours through a potential difference of 40 V.

3

i) $H = I^2 R t$

ii) $H = V.I.t$
 $= V.Q$

Given : $V = 40$ volts , $Q = 96000$ C

$H = 40 \text{ V} \times 96000 \text{ C}$

$= 3.84 \times 10^6 \text{ J}$

1

 $\frac{1}{2}$

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 $\frac{1}{2}$

Why does the cord of an electric oven not glow while its heating element does ?

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Why does the cord of an electric oven not glow while its heating element does ?

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Cord is made up of copper wire whereas heating element is made up of alloy.

1

Define 1 kilowatt hour.

CBSE 2019

Define 1 kilowatt hour.

CBSE 2019

1000 watt of power is used for 1 hour / commercial unit of energy / $3.6 \times 10^6 \text{J}$	1
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- (a) Define electric power. An electrical device of resistance R is connected across a source of voltage V and draws a current I . Derive an expression for power in terms of current and resistance.
- (b) Two electric bulbs rated 100 W; 220 V and 60 W; 220 V are connected in parallel to an electric mains of 220 V. Find the current drawn by the bulbs from the mains.

- (a) Define electric power. An electrical device of resistance R is connected across a source of voltage V and draws a current I . Derive an expression for power in terms of current and resistance.
- (b) Two electric bulbs rated 100 W; 220 V and 60 W; 220 V are connected in parallel to an electric mains of 220 V. Find the current drawn by the bulbs from the mains.

(a) It is the rate at which electrical energy is dissipated or consumed in an electrical circuit is called electric power .

We know

$$V = W/Q$$

$$W = VQ$$

$$W/t = VQ/t$$

$$P = VI$$

$$P = IR.I$$

$$P = I^2R$$

(b) Bulb I: $I = P/V$, 100W 220V

$$I_1 = 100/220A$$

$$= 5/11A$$

Bulb II : 60W 220V

$$I_2 = 60/220A$$

$$= 3/11 A$$

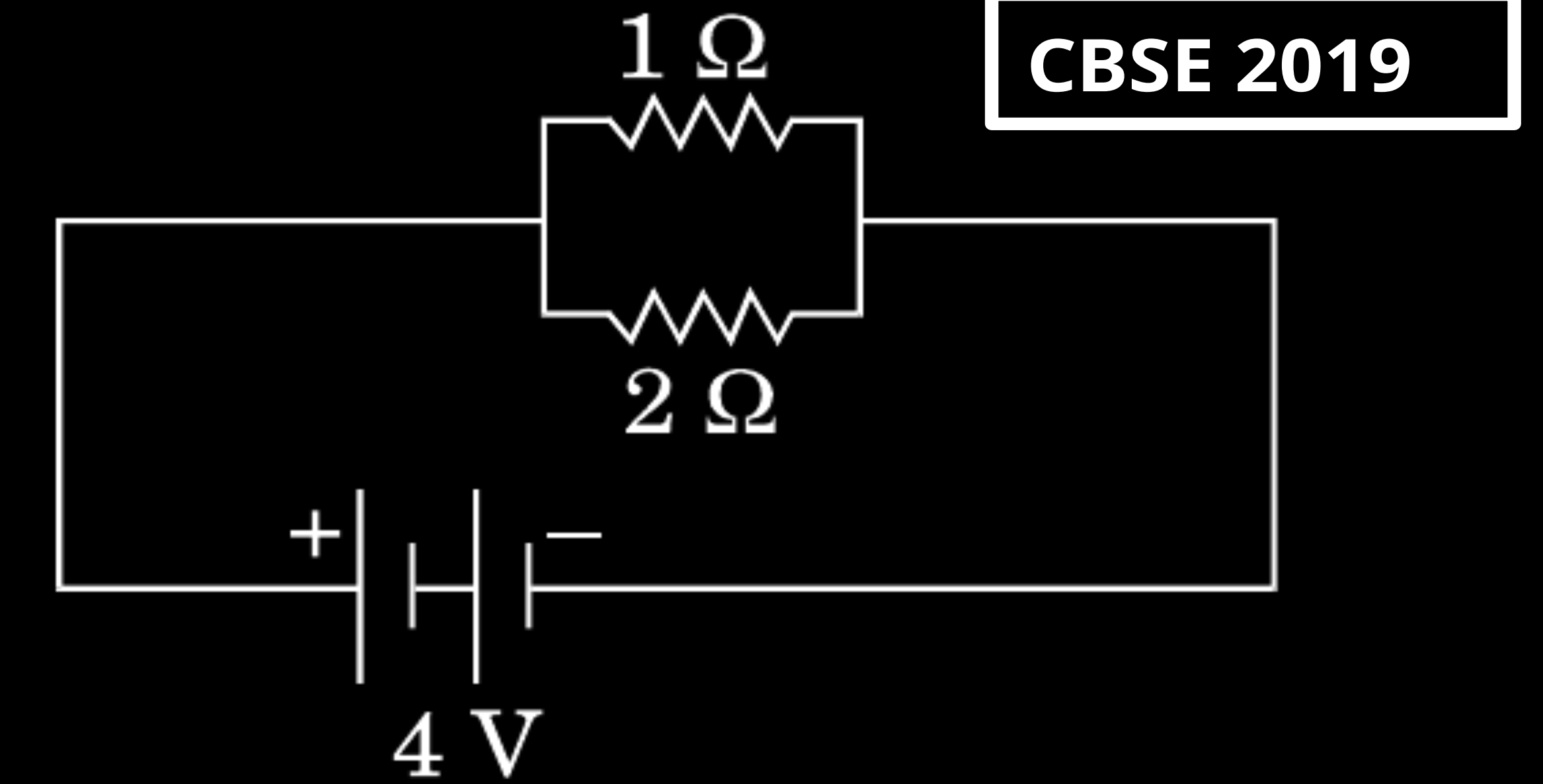
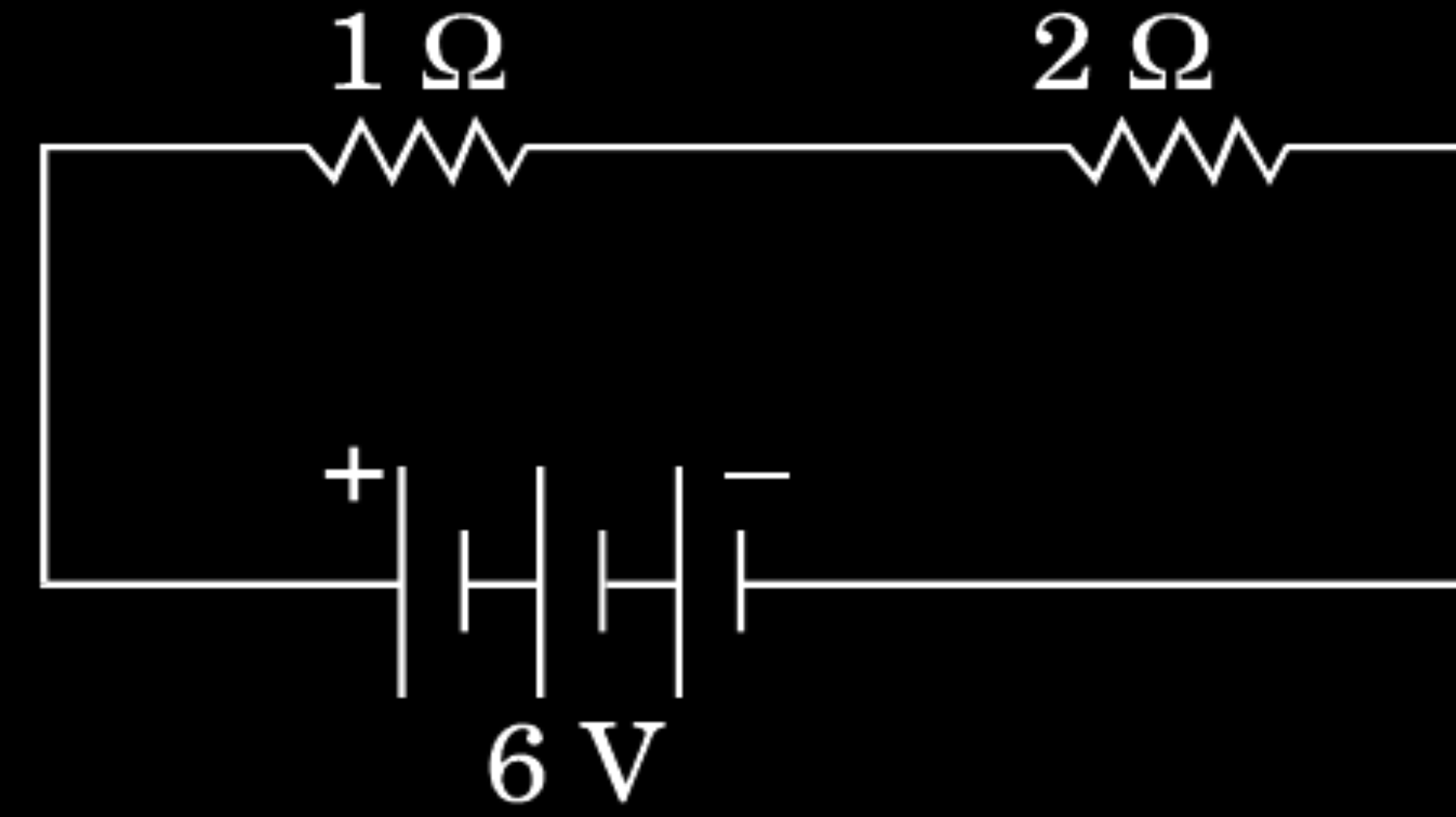
$$\text{Total current } (5/11 + 3/11)A = 8/11A = 0.72A$$

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Compare the power used in $2\ \Omega$ resistor in each of the following circuits :



(A)

- $R = R_1 + R_2$
 $R = 1\Omega + 2\Omega$
 $R = 3\Omega$
- $V = IR$
 $I = V/R$
 $I = \frac{6V}{3\Omega} = 2 \text{ Ampere or } 2 \text{ A}$
- $P = I^2 R$
 $= 2 \times 2 \times 2$
 $= 8 \text{ W}$

(B) $P = V^2/R$
 $P = \frac{4 \times 4}{2}$
 $P = 8 \text{ W}$

$\frac{1}{2}$

$\frac{1}{2}$

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A bulb is rated 40 W; 220 V. Find the current drawn by it, when it is connected to a 220 V supply. Also find its resistance. If the given bulb is replaced by a bulb of rating 25 W; 220 V, will there be any change in the value of current and resistance ? Justify your answer and determine the change.

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$$\begin{aligned} \text{(i)} \quad P &= 40 \text{ W} \\ V &= 220 \text{ V} \\ P &= VI \\ I &= \frac{P}{V} = \frac{40 \text{ W}}{220 \text{ V}} \\ &= 0.18 \text{ A} \end{aligned}$$

1

$$\begin{aligned} \text{(ii)} \quad R &= \frac{V^2}{P} \\ &= \frac{220 \times 220}{40} \\ &= 1210 \Omega \end{aligned}$$

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$$\begin{aligned} \text{(iii)} \quad P &= 25 \text{ W} \\ V &= 220 \text{ V} \\ P &= VI \\ I &= \frac{P}{V} \\ &= \frac{25}{220} = 0.113 \text{ A} \end{aligned}$$

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$$\begin{aligned} \text{(iv)} \quad R &= \frac{V^2}{P} \\ &= \frac{220 \times 220}{25} \\ &= 1936 \Omega \end{aligned}$$

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(v) Yes there is a change in current and resistance

1

A bulb is rated 40 W; 220 V. Find the current drawn by it, when it is connected to a 220 V supply. Also find its resistance. If the given bulb is replaced by a bulb of rating 25 W; 220 V, will there be any change in the value of current and resistance ? Justify your answer and determine the change.

CBSE 2019

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(v) Yes there is a change in current and resistance

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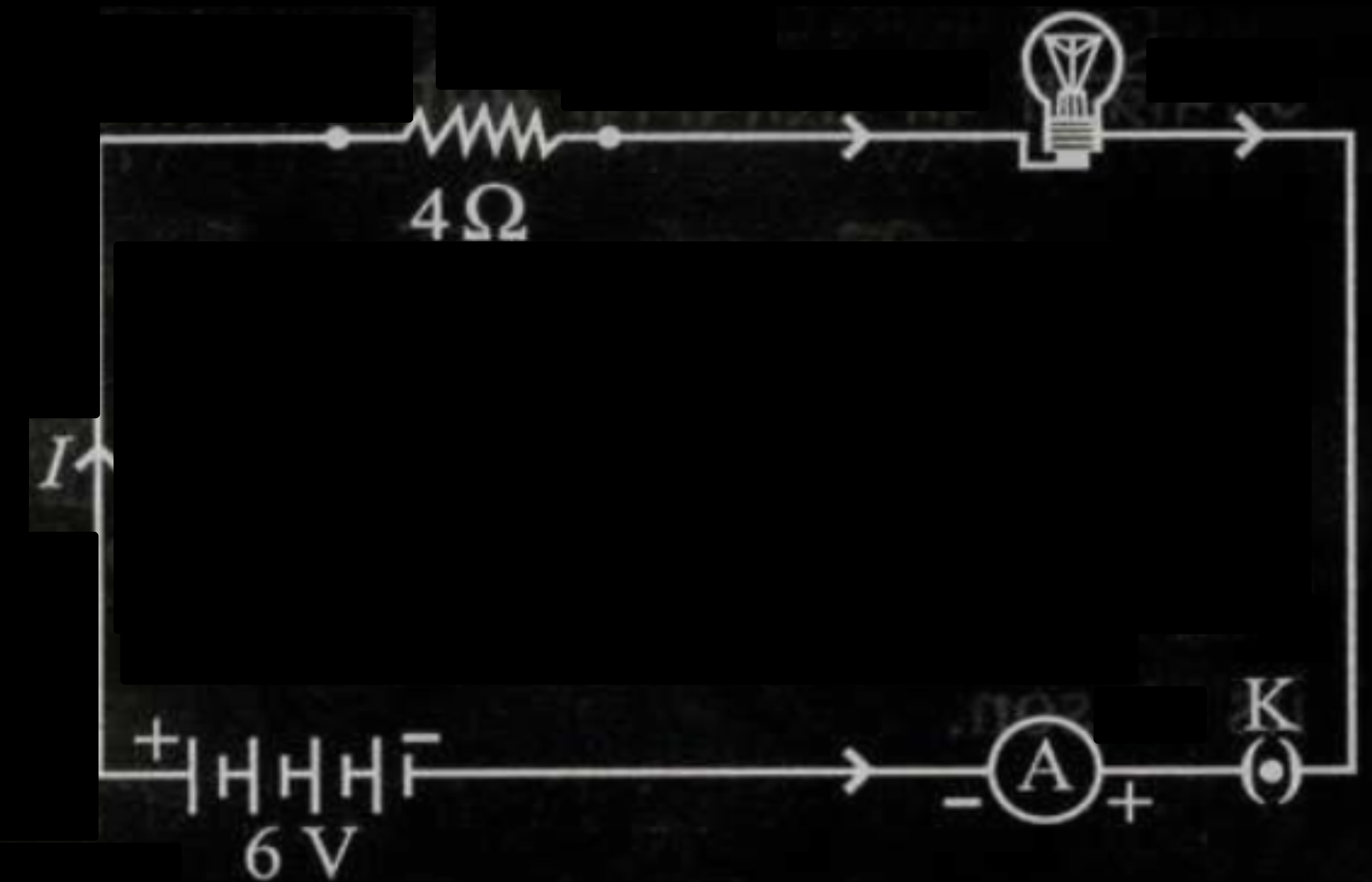
An electric lamp of resistance $20\ \Omega$ and a conductor of resistance $4\ \Omega$ are connected to a $6\ \text{V}$ battery as shown in the circuit. Calculate

(a) the total resistance of the circuit,

(b) the current through the circuit.

(c) the potential difference across the (i) electric lamp and (ii) conductor, and

(d) power of the lamp.



$$\begin{aligned} \text{a)} \quad R &= R_1 + R_2 \\ &= 20 \, \Omega + 4 \, \Omega = 24 \, \Omega \end{aligned}$$

$$\begin{aligned} \text{b)} \quad I &= \frac{V}{R} \\ &= \frac{6V}{24 \, \Omega} = 0.25 \, A \end{aligned}$$

$$\begin{aligned} \text{c) (i) For electric lamp:} \\ V &= IR \\ &= \frac{6}{24} \times 20 = 5 \, V \end{aligned}$$

$$\begin{aligned} \text{(ii) For Conductor:} \\ V &= IR \\ &= \frac{6}{24} \times 4 = 1 \, V \end{aligned}$$

$$\begin{aligned} \text{d) } P &= VI \\ &= 5 \, V \times \frac{6}{24} A = 1.25 \, W \end{aligned}$$

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- (a) Write Joule's law of heating.
- (b) Two lamps, one rated 100 W; 220 V, and the other 60 W; 220 V, are connected in parallel to electric mains supply. Find the current drawn by two bulbs from the line, if the supply voltage is 220 V.

CBSE 2018

- (a) Write Joule's law of heating.
- (b) Two lamps, one rated 100 W; 220 V, and the other 60 W; 220 V, are connected in parallel to electric mains supply. Find the current drawn by two bulbs from the line, if the supply voltage is 220 V.

CBSE 2018

(a) Joule's law of heating – Heat produced in a resistor is (i) directly proportional to the square of current for a given resistance, (ii) directly proportional to the resistance for a given current and (iii) directly proportional to the time for which the current flows through the resistor / $H = I^2Rt$ where, H = Heat produced, I = current, R = Resistance of the conductor and t = Time for which the current flows through the resistor

Note :If the candidate gives only the expression $H = I^2Rt$ award ½ mark only.

(b) Current in 1st bulb, $I_1 = \frac{P_1}{V} = \frac{100}{220} = \frac{5}{11}$ A or 0.45 A
 Current in 2nd bulb, $I_2 = \frac{P_2}{V} = \frac{60}{220} = \frac{3}{11}$ A or 0.27 A

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